第一章 光的干涉



§1-1 相干光

§1-2 双缝干涉

§1-3 光程与光程差

§1-4 薄膜干涉

§1-5 迈克耳孙干涉仪

§1-6 多光束干涉

(和衍射部分数学类似，不做要求)

来源：

程守洙 普通物理学

课件 （有改动）

从Maxwell方程组开始

In general:

$$
\begin{cases}
\nabla \cdot \mathbf{E} = \dfrac{1}{\epsilon_0}\rho \\
\nabla \times \mathbf{E} = -\dfrac{\partial \mathbf{B}}{\partial t} \\
\nabla \cdot \mathbf{B} = 0 \\
\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \dfrac{\partial \mathbf{E}}{\partial t}
\end{cases}
$$

光源：发光的物体

类型：·普通光源(自发辐射)

·激光光源(受激辐射)

热（辐射）光源

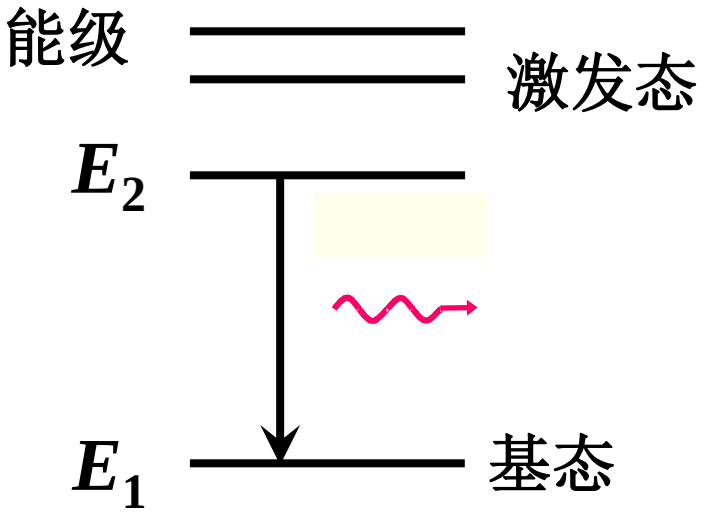
非热辐射光源（冷光源）

气体放电管、日光灯、萤火虫……

§1-1 相干光

一、普通光源的发光机理

原子的发光模型：



激发态寿命 10^{-11}～10^{-8} s

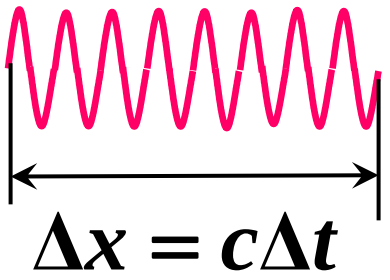
自发辐射产生一定频率、振动方向，长度有限的光波列

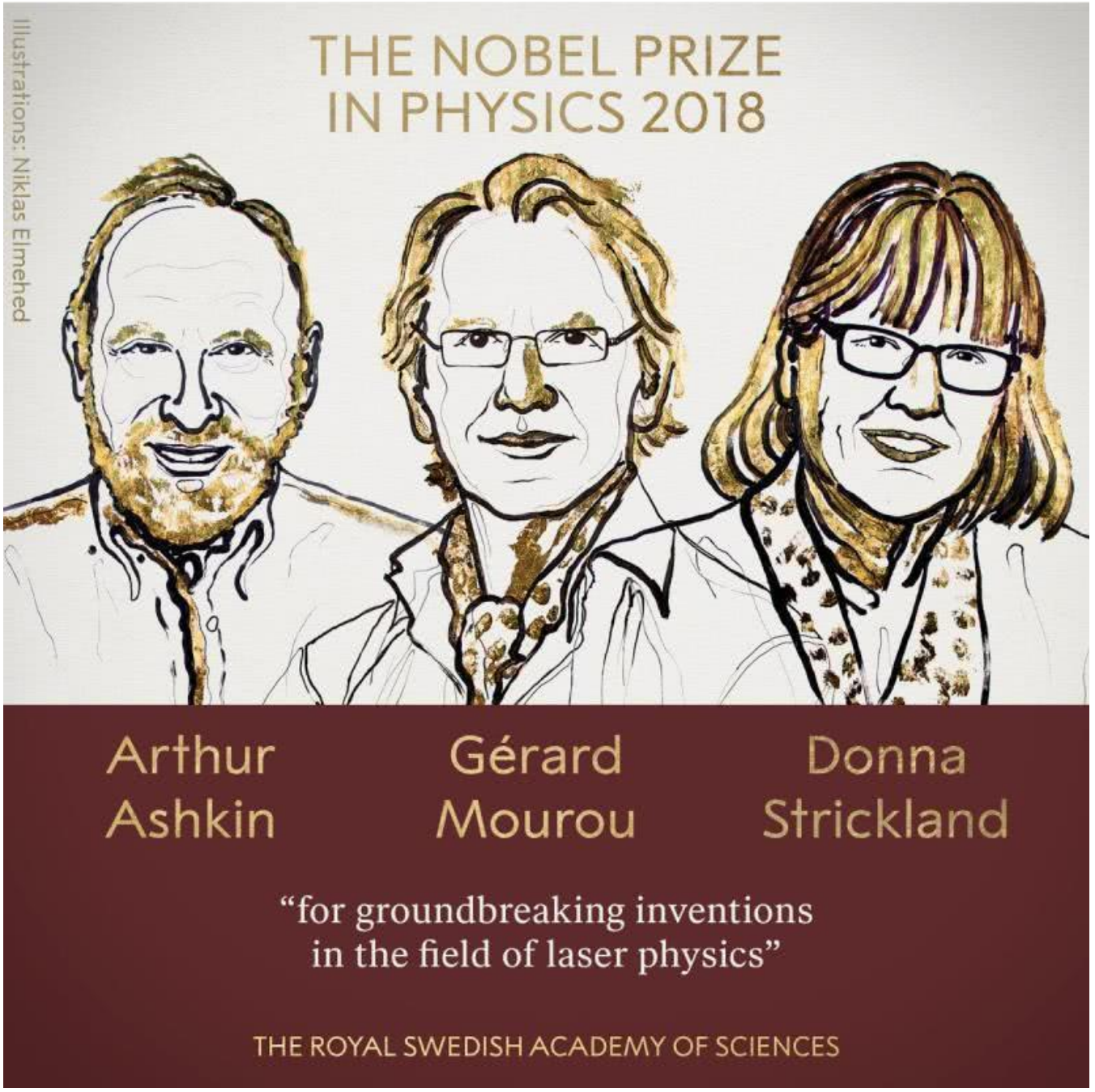
普通光源如白炽灯、太阳……

大量原子和分子持续、随机地发射的光波列

两个特点：间歇性、随机性

原子光波列




THE NOBEL PRIZE
IN PHYSICS 2018
Illustrations: Niklas Elmehed
Arthur Ashkin
Gérard Mourou
Donna Strickland
"for groundbreaking inventions
in the field of laser physics"
THE ROYAL SWEDISH ACADEMY OF SCIENCES

THE NOBEL PRIZE IN PHYSICS 2023

Illustrations: Niklas Elmehed



Pierre Agostini

Ferenc Krausz

Anne L'Huillier

"for experimental methods that generate attosecond pulses of light for the study of electron dynamics in matter"

THE ROYAL SWEDISH ACADEMY OF SCIENCES

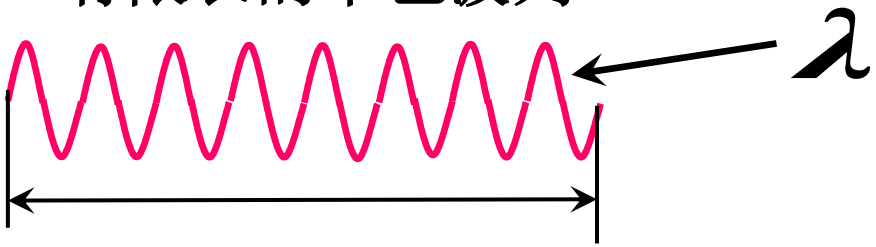
二、单色光

可见光波长 400—700nm

单色光：具有单一频率或波长

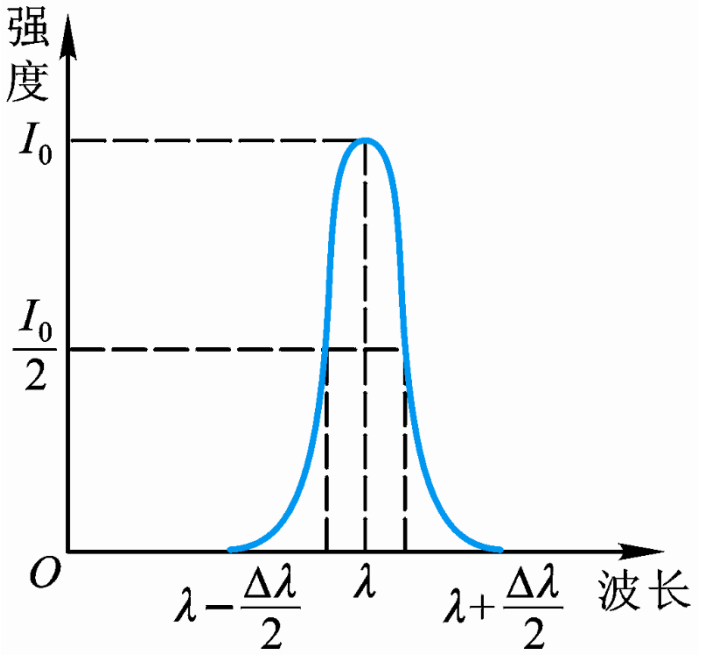
理想单色光　　$\Delta\nu \to 0, \Delta t \to \infty$　或　$\Delta\lambda \to 0, \Delta x \to \infty$

有限长的单色波列



$$\Delta x = c\Delta t$$

$\Delta\lambda$ ：谱线宽度



普通单色光源：　$\Delta\lambda < 0.1 \sim 10^{-3}\text{nm}$

激光：　　$\Delta\lambda < 10^{-9}\text{nm}$

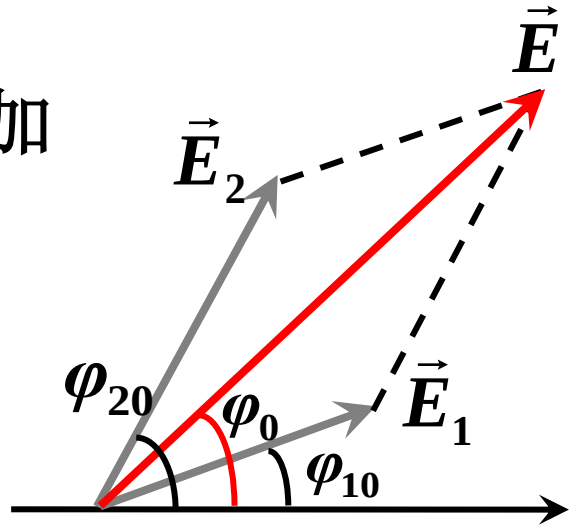
三、相干光

光波中的电场矢量 $\vec{E}$ 称为光矢量。

两振动方向相同的同频单色光的叠加

$$E_1 = E_{10}\cos(\omega t + \varphi_{10})$$

$$E_2 = E_{20}\cos(\omega t + \varphi_{20})$$



$$E = E_1 + E_2 = E_0\cos(\omega t + \varphi_0)$$

$$E_0 = \sqrt{E_{10}^2 + E_{20}^2 + 2E_{10}E_{20}\cos\Delta\varphi}$$

$$\Delta\varphi = \varphi_2 - \varphi_1$$

注意：这里的相位没有加角标0

平均光强： $I = I_1 + I_2 + 2\sqrt{I_1 I_2}\ \overline{\cos\Delta\varphi}$

对于普通光源发出的光 $\overline{\cos\Delta\varphi} = 0$

$\Rightarrow$ $I = I_1 + I_2$ 非相干叠加

如果 $\Delta\varphi = \varphi_2 - \varphi_1 =$ 常量

$\Rightarrow$ $I = I_1 + I_2 + 2\sqrt{I_1 I_2}\ \cos\Delta\varphi$ 相干叠加

若两个光源发出的光之间具有确定的相位差，则把这两个光源称为相干光源，它们所发出的就是相干光。

相干条件：振动方向相同，频率相同，具有确定的相位差。

四、相干光的获得方法

把由同一光源上同一点发出的光一分为二，经由不同的路径后再使之叠加起来

分波阵面法：如杨氏双缝实验

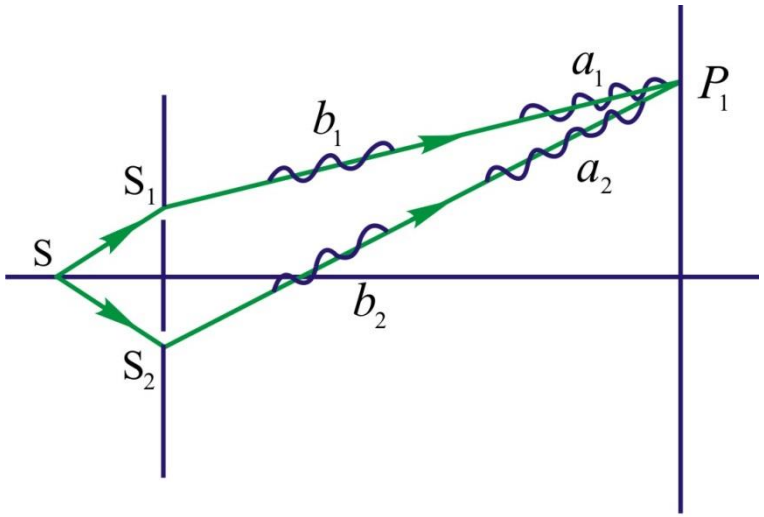
分振幅法：如薄膜干涉实验

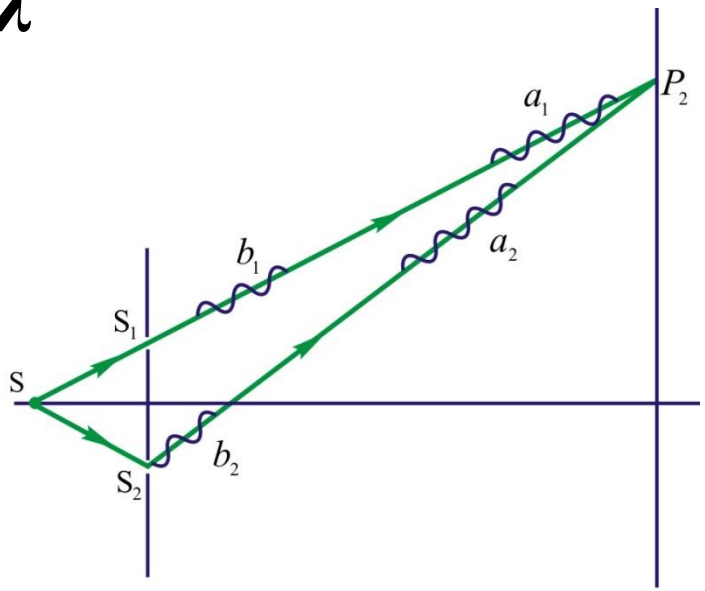
现在， 利用高度相干性的激光作为相干光源

五、光源的相干长度

根据普通光源发光的特点，在干涉装置中分出来的两束光会聚时，波程差不能大于原子光波列的长度——相干长度。

相干长度：$L = \delta_{\max} = \dfrac{\lambda^2}{\Delta\lambda}$　　$\delta > L$ 则无干涉





光源单色性越好，相干长度越长，其相干性越好。

如激光光源，相干长度约为几十到几百米，相干性极好。

不确定性原理

- 光源发光时间和波包的频率范围满足：

$$\Delta v \Delta t \approx 1$$

- 不确定性原理由Fourier 变换导出，因此它适用于任何波：机械波，电磁波，物质波等
- 或者，相干长度理解为空间上的拍

The superposition of many waves

$$E_0^2 = \sum_{i=1}^{N} E_{0i}^2 + 2 \sum_{j>i}^{N} \sum_{i=1}^{N} E_{0i} E_{0j} \cos(\alpha_i - \alpha_j)$$

and

$$\tan\alpha = \frac{\sum_{i=1}^{N} E_{0i} \sin\alpha_i}{\sum_{i=1}^{N} E_{0i} \cos\alpha_i}$$

- The addition of N harmonic waves: incoherent

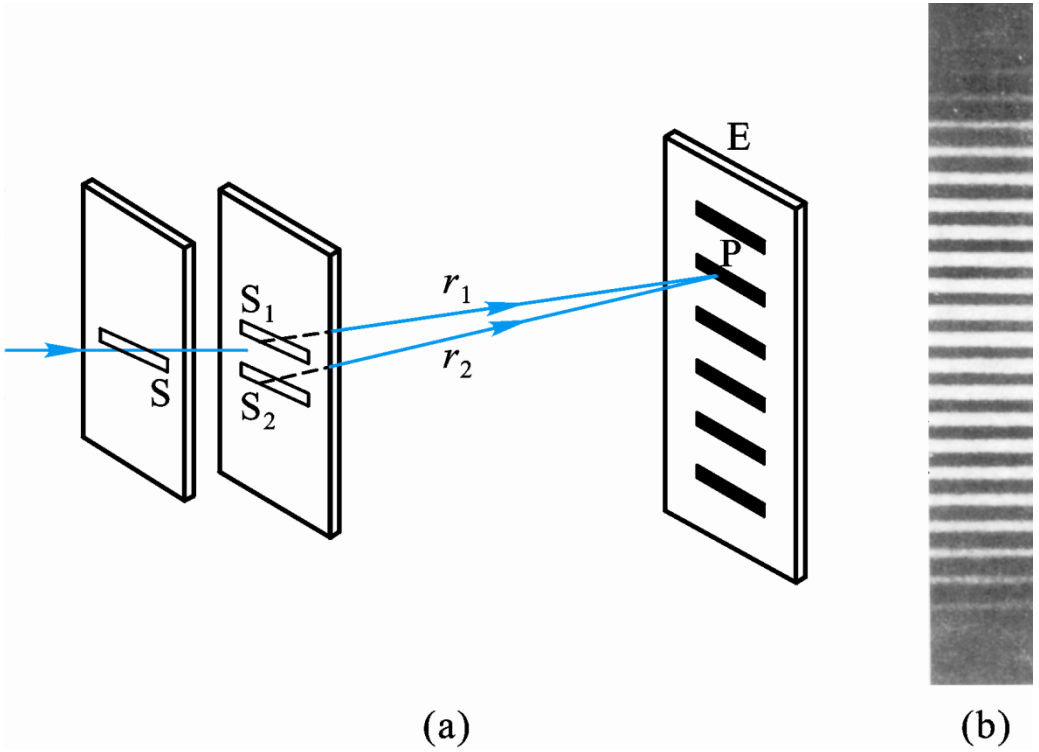
Summation of many atoms and many lamps.

- The addition of N vectors→random walk,

Brown motion and existence of atoms

§1-2 双缝干涉

一、杨氏双缝实验

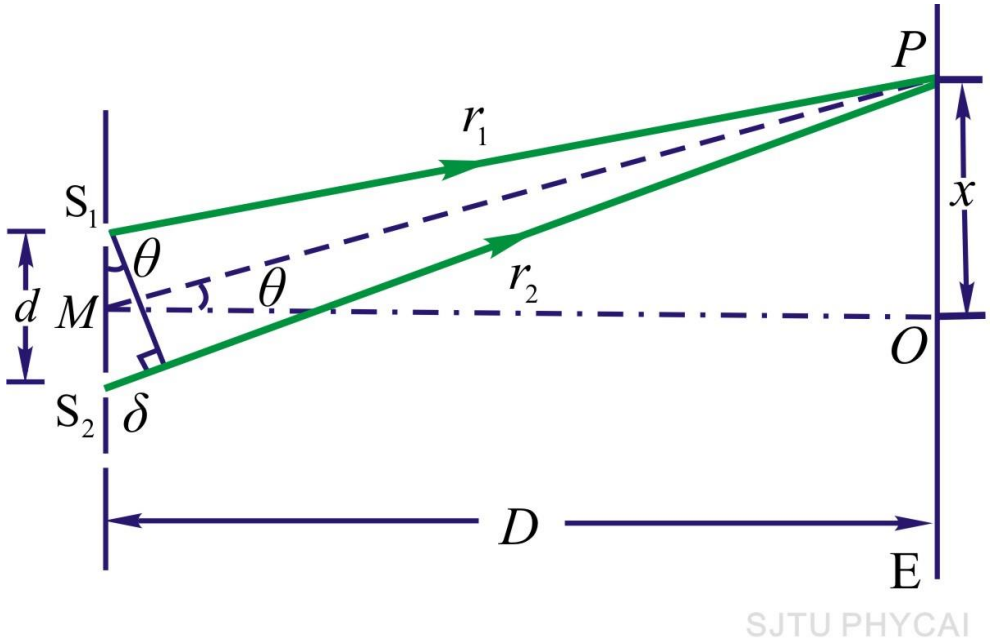


波阵面分割 ⇨ 相干光

杨氏实验为光的波动理论奠定了实验基础

二、干涉明暗条纹的位置

$$D >> d, D >> x$$



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波程差：

$$\delta = r_2 - r_1 \approx d\sin\theta$$

$$\approx d\ \tan\theta = d\cdot\frac{x}{D}$$

明纹 $\delta = \pm k\lambda$ $x_{\pm k} = \pm k\dfrac{D}{d}\lambda$ （$k = 0,1\ 2,\cdots$）

暗纹 $\delta = \pm(2k+1)\dfrac{\lambda}{2}$ $x_{\pm(2k+1)} = \pm(2k+1)\dfrac{D}{2d}\lambda$

条纹间距 $\Delta x = x_{k+1} - x_{\pm k} = \dfrac{D}{d}\lambda$

明纹　$\delta = \pm k\lambda$　$x_{\pm k} = \pm k\dfrac{D}{d}\lambda$　（$k = 0,1,2,\cdots$）

暗纹　$\delta = \pm(2k+1)\dfrac{\lambda}{2}$　$x_{\pm(2k+1)} = \pm(2k+1)\dfrac{D}{2d}\lambda$

条纹间距　$\Delta x = x_{k+1} - x_{\pm k} = \dfrac{D}{d}\lambda$

条纹特点：

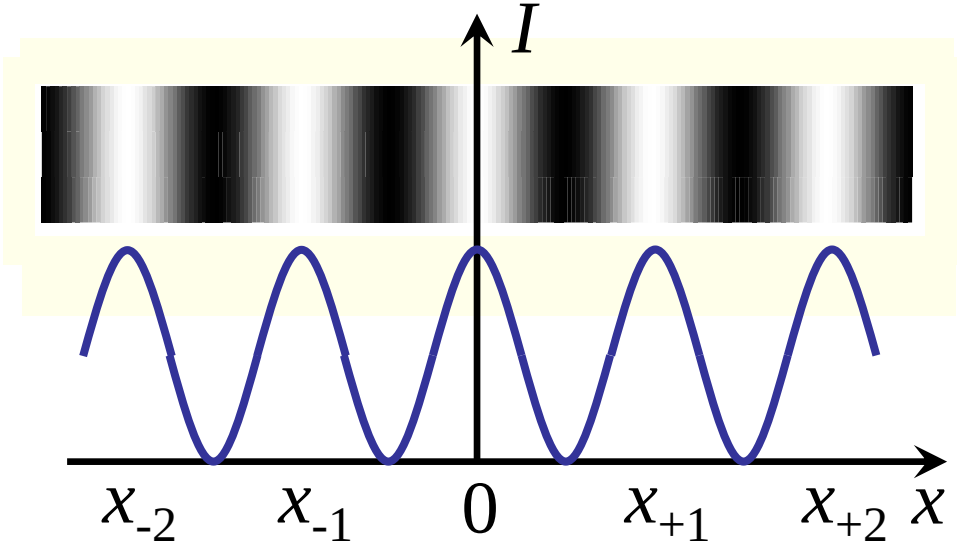
（1）平行的明暗相间条纹

（2）条纹等间距

（3）中间级次低

（4）$\Delta x \propto \dfrac{1}{d}$

x、$\Delta x \propto \lambda$



三、干涉条纹的强度分析

当两束相干光在空间任一点相遇时，它们之间的相位差$\Delta\varphi$随空间位置不同而连续变化，从而在不同位置上出现光强的强弱分布。

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\ \cos\Delta\varphi \qquad \Delta\varphi = \frac{2\pi\delta}{\lambda}$$

干涉相长（明） $\Delta\varphi = \pm 2k\pi, \quad (k = 0,1,2,3,\ldots)$

$$I = I_{\max} = I_1 + I_2 + 2\sqrt{I_1 I_2}$$

干涉相消（暗） $\Delta\varphi = \pm(2k+1)\pi, \quad (k = 0,1,2,3,\ldots)$

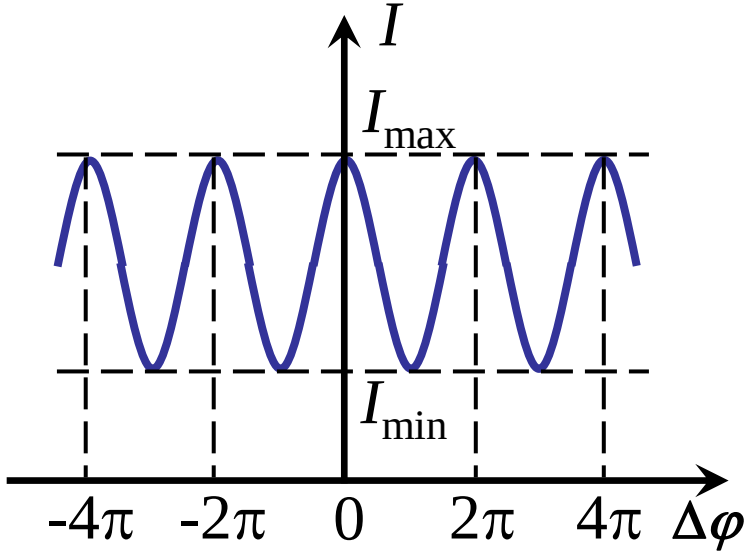
$$I = I_{\min} = I_1 + I_2 - 2\sqrt{I_1 I_2}$$

干涉的光强分布

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\ \cos\Delta\varphi$$

$$I = I_{max} = I_1 + I_2 + 2\sqrt{I_1 I_2}$$

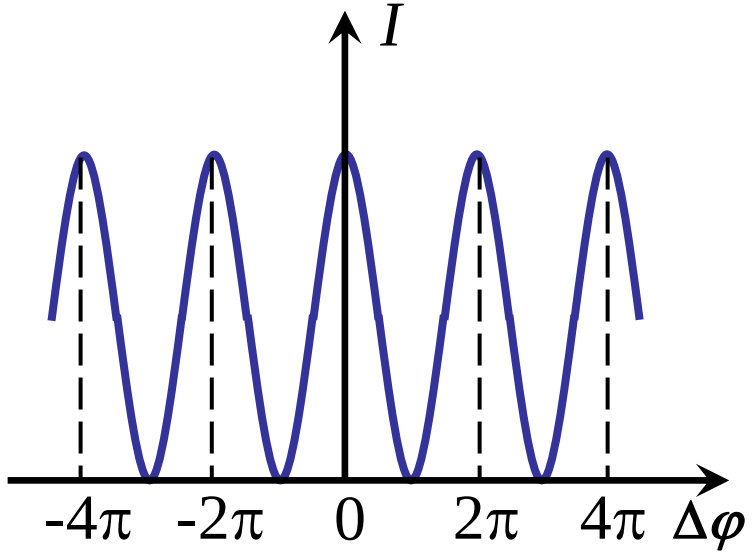
$$I = I_{min} = I_1 + I_2 - 2\sqrt{I_1 I_2}$$



若 $I_1 = I_2 = I_0$

$$I = 4I_0 \cos^2 \frac{\Delta\varphi}{2}$$

$$I_{max} = 4I_0 \qquad I_{min} = 0$$



例 在杨氏双缝实验中，屏与双缝间的距离 D =1m，用钠光灯作单色光源（λ =598.3 nm），问：（1） d =2 mm 和 d =10 mm两种情况下，相邻明纹间距各为多大？（2）如肉眼仅能分辨两条纹的间距为0.15 mm，现用肉眼观察干涉条纹，问双缝的最大间距是多少？

解： （1）相邻两明纹间的距离为

$$\Delta x = \frac{D\lambda}{d}$$

d =2 mm时， $\Delta x = \dfrac{1\times 589.3\times 10^{-9}}{2\times 10^{-3}} = 0.295\,(\mathrm{mm})$

d =10 mm时， $\Delta x = \dfrac{1\times 589.3\times 10^{-9}}{10\times 10^{-3}} = 0.059\,(\mathrm{mm})$

（2）如Δx =0.15 mm，则

$$d = \frac{D\lambda}{\Delta x} = \frac{1\times 589.3\times 10^{-9}}{0.15\times 10^{-3}} \approx 4(\text{mm})$$

在这样的条件下，双缝间距必须小于4 mm才能看到干涉条纹。

例 在杨氏实验装置中，采用加有蓝绿色滤光片的白色光源，波长范围$\Delta\lambda$ =100 nm，平均波长为490 nm。试估算从第几级开始，条纹条纹将变得无法分辨？

解： 设该蓝绿光的波长范围为$\lambda_1 \to \lambda_2$，则有

$$\lambda_2 - \lambda_1 = \Delta\lambda = 100\,\text{nm} \qquad \frac{1}{2}(\lambda_1 + \lambda_2) = 490\,\text{nm}$$

相应于λ_1 和λ_2，杨氏干涉条纹中k 级明纹的位置分别为

$$x_1 = k\frac{D}{d}\lambda_1 \qquad x_2 = k\frac{D}{d}\lambda_2$$

k 级条纹所占的宽度为

$$x_2 - x_1 = k\frac{D}{d}\lambda_2 - k\frac{D}{d}\lambda_1 = k\frac{D}{d}\Delta\lambda$$

当此宽度大于或等于相应于平均波长 $\bar{\lambda}$ 的条纹间距时，干涉条纹变得模糊不清，即

$$k\frac{D}{d}\Delta\lambda \geq \frac{D}{d}\bar{\lambda}$$

$$\Rightarrow \quad k \geq \frac{\bar{\lambda}}{\Delta\lambda} = 4.9$$

从第五级开始，干涉条纹变得无法分辨。

一般而言：可得光的最大相干光程差

$$\Delta l_{\mathrm{M}} = \frac{\lambda^2}{\Delta\lambda}$$

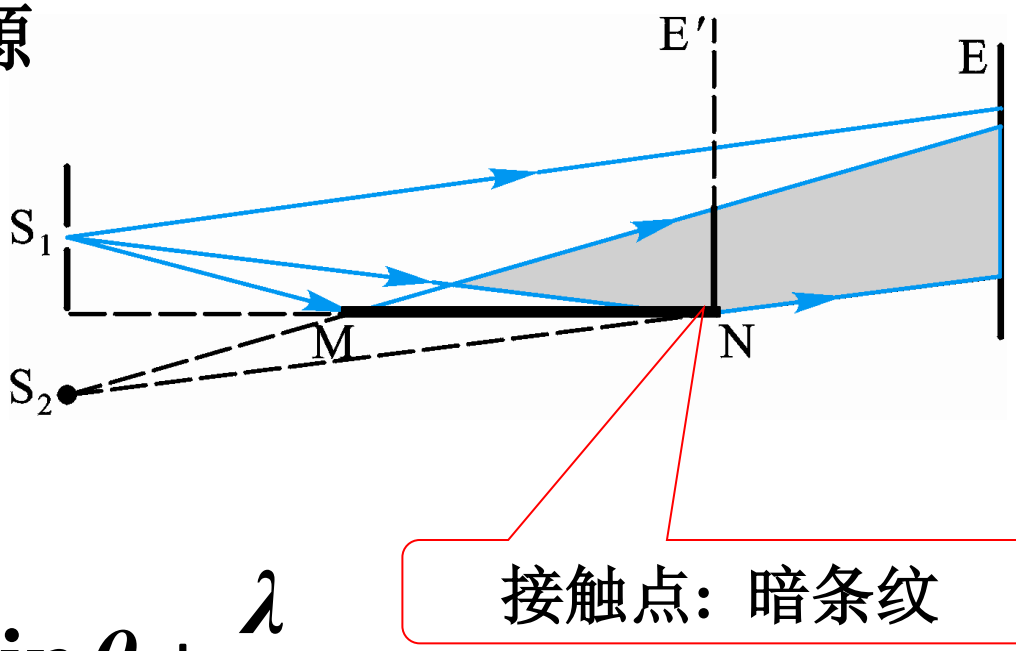
四、 洛埃德镜实验

S_1：实光源，S_2：虚光源

MN：平面反射镜

阴影部分出现干涉

反射光存在半波损失。



$$\delta = r_2 - r_1 + \frac{\lambda}{2} = d\sin\theta + \frac{\lambda}{2}$$

半波损失：光从光疏介质到光密介质的分界面上反射时，在垂直入射（i =0°）或掠入射（i =90° ）的情况下，反射光的相位与入射光的相位有 π 的突变。

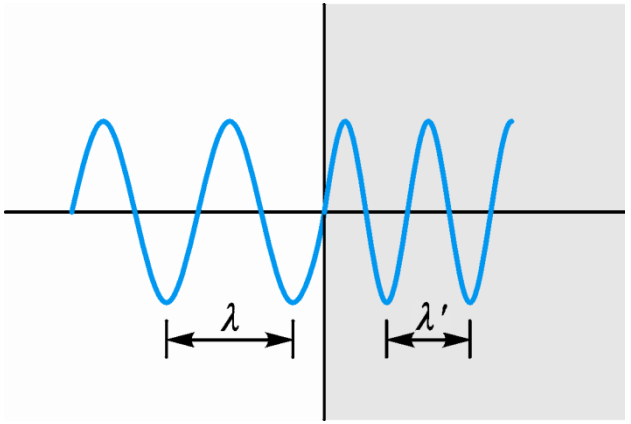
§1-3 光程与光程差

一、光程

单色光（ν）在真空中的波长：$\lambda = \dfrac{c}{\nu}$

在介质（折射率为 n）中的波长：

$$\lambda' = \frac{u}{\nu} = \frac{c/n}{\nu} = \frac{c/\nu}{n} = \frac{\lambda}{n}$$

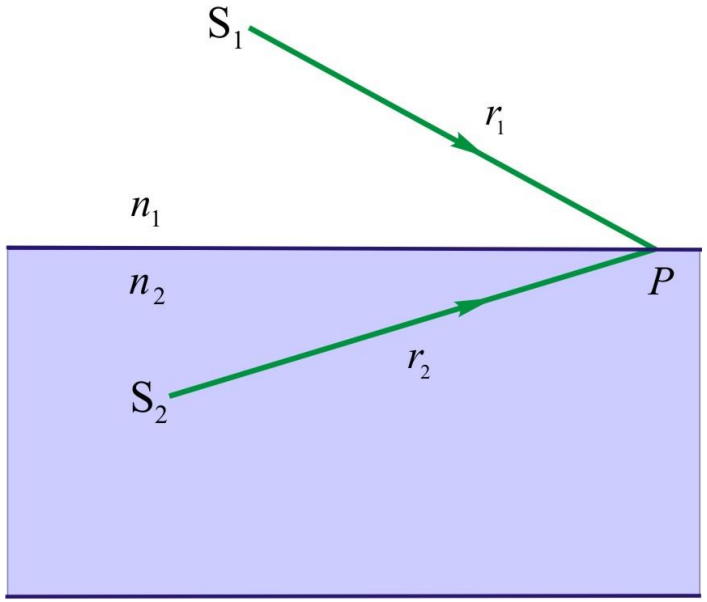


光程（nx）：将光在介质中通过的路程折算到同一时间内在真空中所通过的相应路程。

$$\frac{x}{\lambda'}\lambda = nx$$

二、光程差

例：两相干波源S_1、S_2产生相干波叠加



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$$\Delta\varphi = \frac{2\pi r_2}{\lambda_2} - \frac{2\pi r_1}{\lambda_1}$$

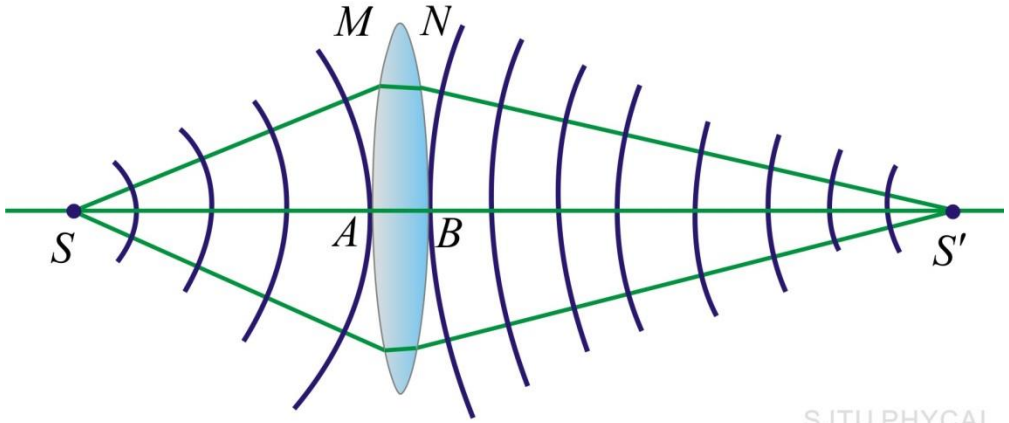
$$= \frac{2\pi n_2 r_2}{\lambda} - \frac{2\pi n_1 r_1}{\lambda} = \frac{2\pi}{\lambda}(n_2 r_2 - n_1 r_1)$$

光程差：$\delta = n_2 r_2 - n_1 r_1$

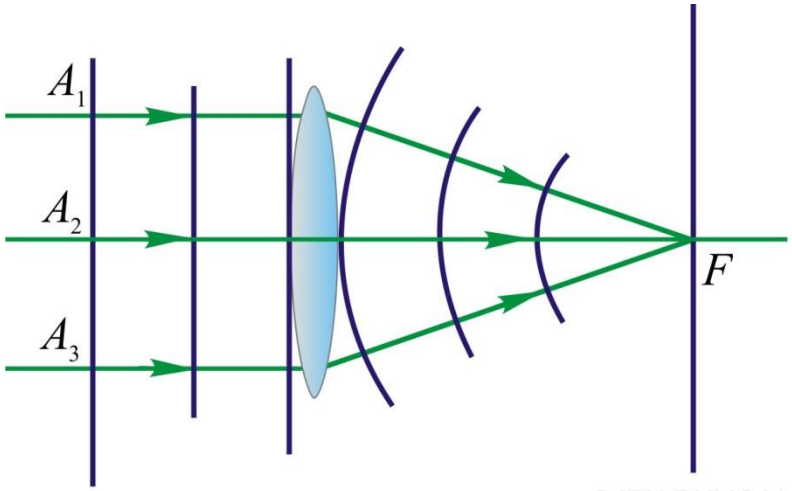
相位差和光程差的关系：$\Delta\varphi = \frac{\delta}{\lambda} 2\pi$

三、物像之间的等光程性

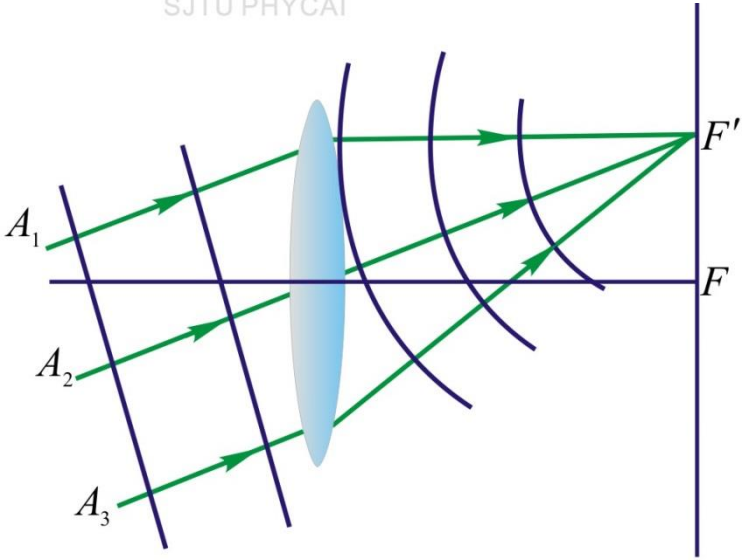
光路中插入薄透镜不会产生附加的光程差。



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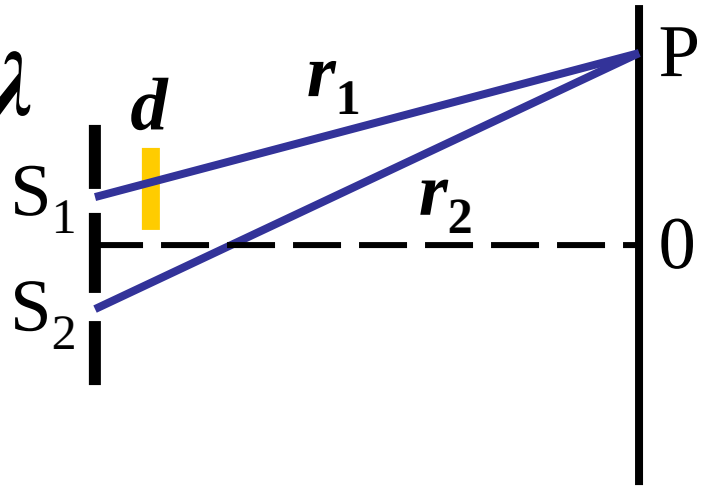


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思考： 用薄云母片（n=1.58）覆盖在杨氏双缝的一条缝上，干涉条纹如何移动？如果这时屏上的零级明纹移到原来的第七级明纹处, 已知入射光波长为550 nm，问云母片的厚度为多少？

整套条纹向上移动

P 点处原来光程差 $r_2 - r_1 = 7\lambda$



插入云母后，P点处光程差

$$r_2 - (r_1 - d + nd) = 0$$

$$\Rightarrow \quad d = \frac{7\lambda}{n-1} = 6.6\times10^{-6}\,\mathrm{m}$$

四、反射光的相位突变和附加光程差

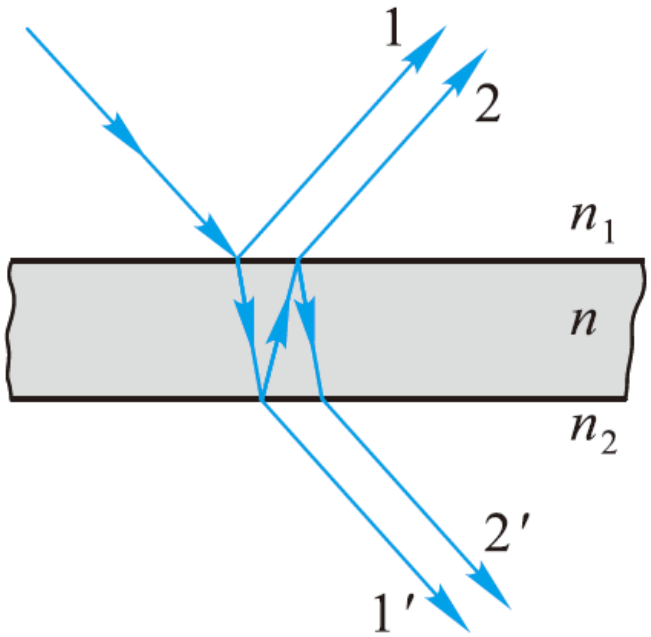
半波损失：光从光疏介质到光密介质的分界面上反射时，反射光有 π 相位突变

相当于一个附加光程差：$\delta = \frac{\lambda}{2}$

对于薄膜：

上下表面反射的两束光发生附加光程差$\lambda/2$的条件：

$n_1 < n > n_2$ 或 $n_1 > n < n_2$

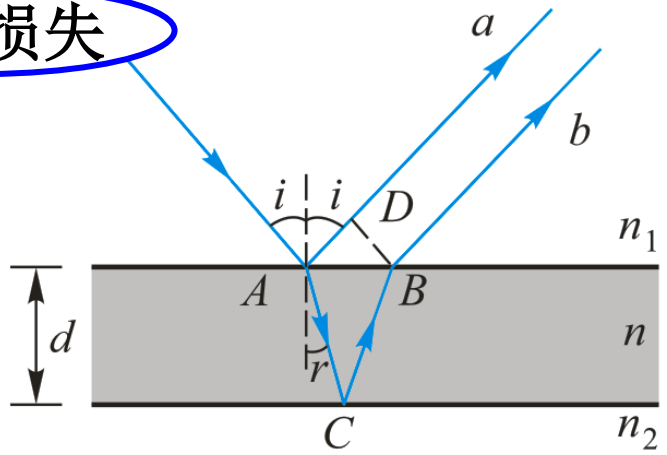


折射光在任何情况下都不会有相位突变。

例如图折射率为n厚度为d的均匀薄膜，一光线以入射角i射到薄膜，在其上下表面反射后，又折回膜的上方成为光线a和b，求它们的光程差

解：

半波损失



$$\delta = n(AC + BC) - n_1 AD + \frac{\lambda}{2}$$

$$AC = BC = \frac{d}{\cos r}$$

$$AD = AB\sin i = 2d \cdot \tan r \cdot \sin i$$

$$\delta = 2n\frac{d}{\cos r} - 2n_1 d \cdot \tan r \cdot \sin i + \frac{\lambda}{2}$$

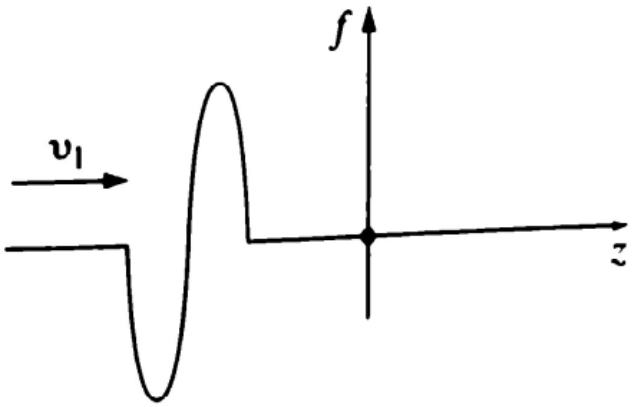
$$n_1 \sin i = n \sin r$$

$$\Rightarrow \quad \delta = 2d\sqrt{n^2 - n_1^2 \sin^2 i} + \frac{\lambda}{2}$$

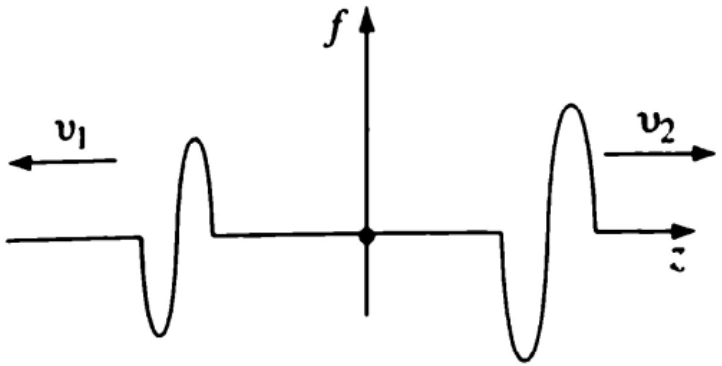
从波的传播的角度理解半波损失

Amplitude between incident, reflected, and transmited Waves (from Griffiths):

$$A_R = \left(\frac{v_2 - v_1}{v_2 + v_1}\right) A_I, \quad A_T = \left(\frac{2v_2}{v_2 + v_1}\right) A_I.$$



(a) Incident pulse



(b) Reflected and transmitted pulses

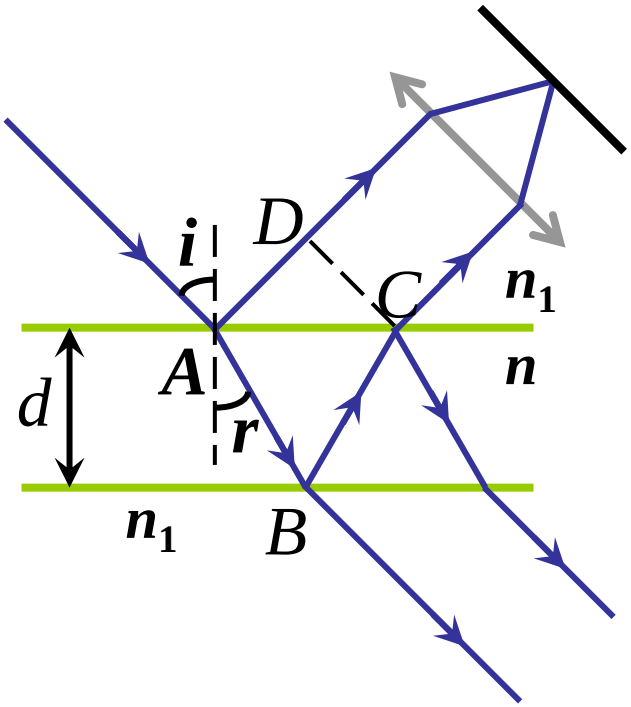
§1-4 薄膜干涉

光经薄膜上下两表面反射后相互叠加所形成的干涉现象，称为薄膜干涉。—— 振幅分割法

可分成等倾干涉和等厚干涉两类。

一、等倾干涉条纹

点光源照射到表面平整，厚度均匀的薄膜上产生的干涉条纹，称等倾干涉条纹。

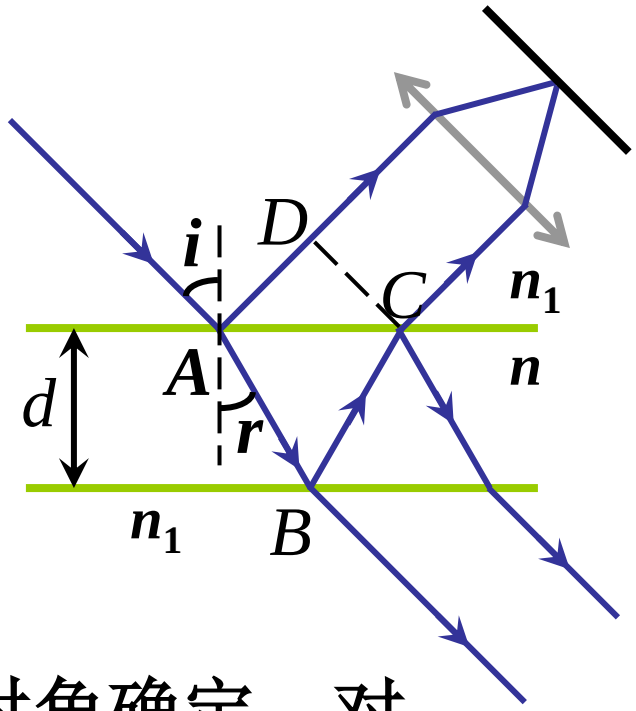


考虑两反射光：

$$\delta = 2d\sqrt{n^2 - n_1^2 \sin^2 i} + \frac{\lambda}{2}$$

$$\delta = 2d\sqrt{n^2 - n_1^2 \sin^2 i} + \frac{\lambda}{2}$$

$$\delta = \begin{cases} k\lambda & \text{明纹} \\ (2k+1)\dfrac{\lambda}{2} & \text{暗纹} \end{cases}$$



讨论

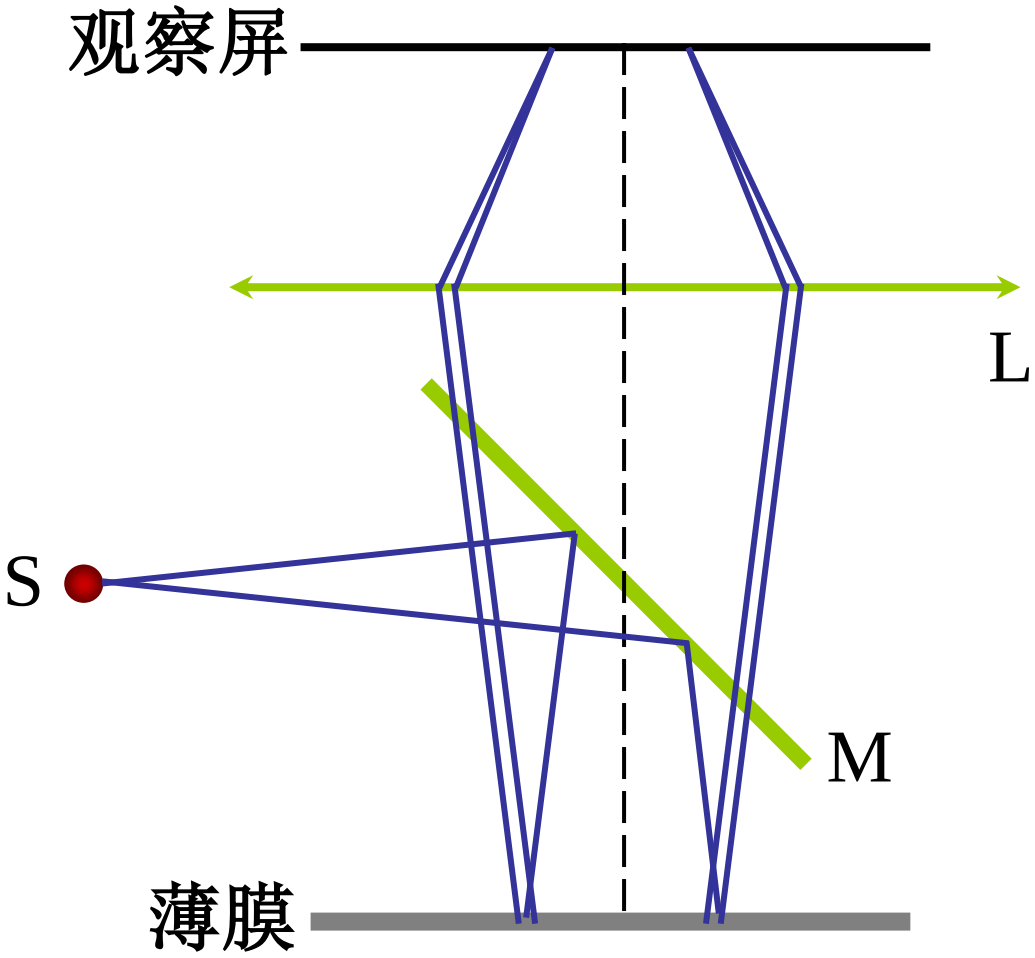
1.由于薄膜厚度均匀，光程差由入射角确定，对于同一级条纹具有相同的倾角，因此称为等倾干涉条纹。

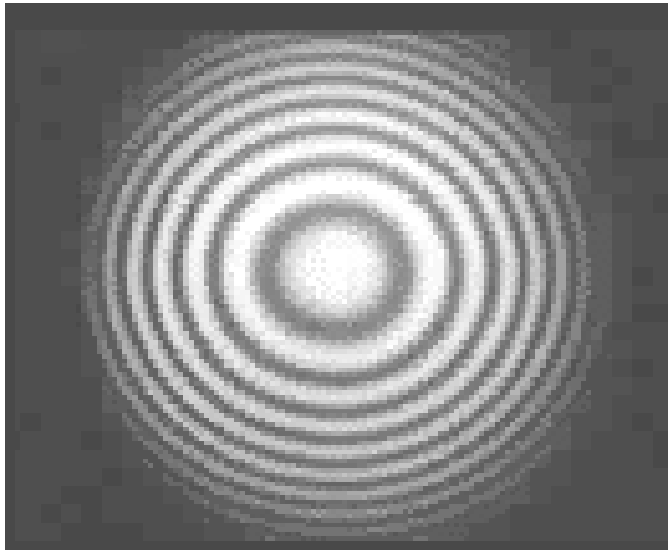
2.对于两透射光，$\delta = 2d\sqrt{n_2^2 - n_1^2 \sin^2 i}$

与反射光的干涉条纹形成互补。

3.是否要考虑半波损失看具体情况而定。

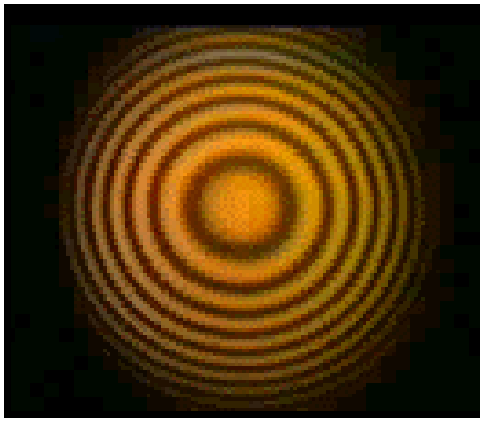
等倾干涉条纹的实验装置





- 条纹特点

$$\delta = 2d\sqrt{n_2^2 - n_1^2 \sin^2 i} + \frac{\lambda}{2} = k\lambda$$



形状： 内疏外密同心圆环

条纹级次分布：

d 一定时，$k \uparrow \rightarrow \delta \uparrow \rightarrow i \downarrow \rightarrow R_k \downarrow$ 中央级次高

若膜厚增大， k一定时， $d \uparrow \rightarrow i \uparrow \rightarrow R_k \uparrow$

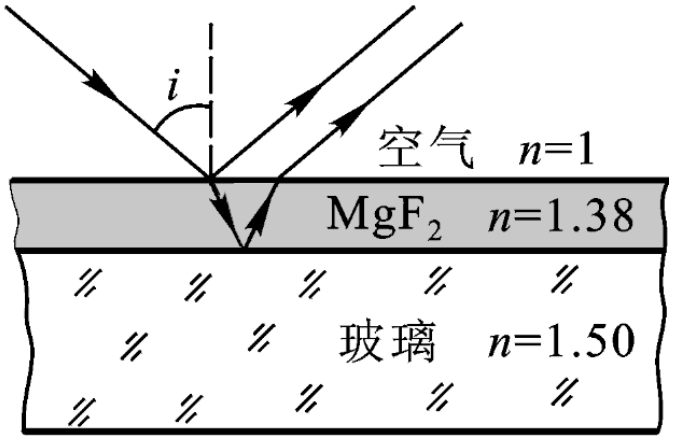
条纹由中央冒出

波长对条纹的影响：

k，d一定时， $\lambda \uparrow \rightarrow i \downarrow \rightarrow R_k \downarrow$

二、增透膜和高反射膜

利用薄膜干涉使反射光减小，这样的薄膜称为增透膜。



上下表面反射都有相位突变π

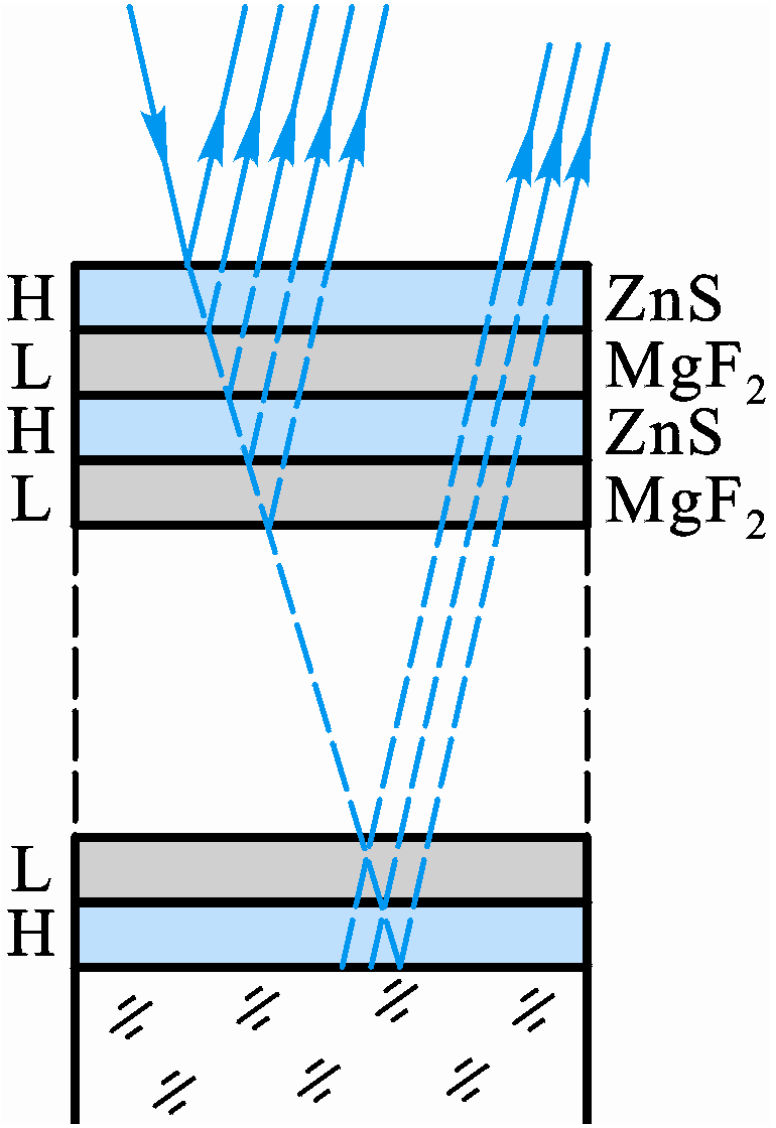
$$2nd=(k+\frac{1}{2})\lambda \quad k=0,1,2,\cdots$$

增透膜的最小厚度：

$$d=\frac{\lambda}{4n}$$

高反射膜(HLHLH…)

反射镜表面交替镀上光学厚度均为$\lambda/4$的高折射率ZnS膜和低折射率的MgF_2膜，形成多层高反射膜。



三、等厚干涉条纹

平行光照射到表面平整、厚度不均匀的薄膜上产生的干涉条纹。

$$\delta = 2d\sqrt{n_2^2 - n_1^2 \sin^2 i} + \frac{\lambda}{2}$$

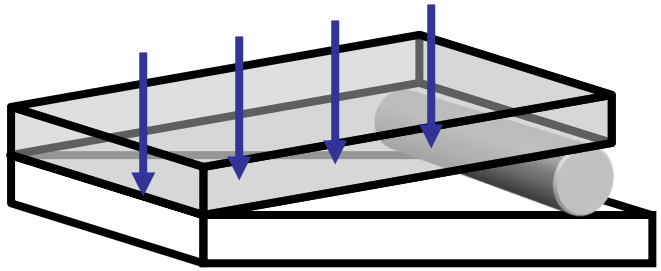
薄膜厚度d 相同之处对应于同一级条纹，因此称为等厚干涉条纹。

光线垂直入射时：

$$\delta = 2nd + \frac{\lambda}{2} = \begin{cases} k\lambda & k = 1,2,\cdots \quad \text{明纹} \\ (2k+1)\dfrac{\lambda}{2} & k = 0,1,2,\cdots \quad \text{暗纹} \end{cases}$$

1、劈尖膜

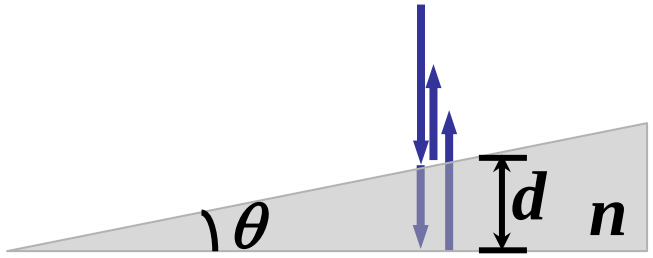
平行光垂直入射，薄膜上下表面之间夹角极小。



$$\delta = 2nd + \frac{\lambda}{2} = \begin{cases} k\lambda & k = 1,2,\cdots \quad \text{明纹} \\ (2k+1)\dfrac{\lambda}{2} & k = 0,1,2,\cdots \quad \text{暗纹} \end{cases}$$

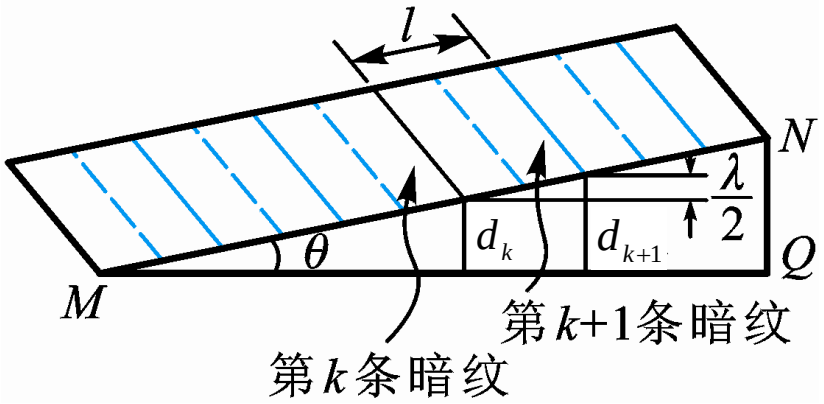
条纹特点：

平行于劈尖棱边等间距的直条纹



接触处为暗条纹（半波损失）

$$\delta = 2nd + \frac{\lambda}{2} = k\lambda$$



相邻条纹所对应的厚度差：

$$d_{k+1} - d_k = \frac{\lambda}{2n}$$

相邻条纹间距：

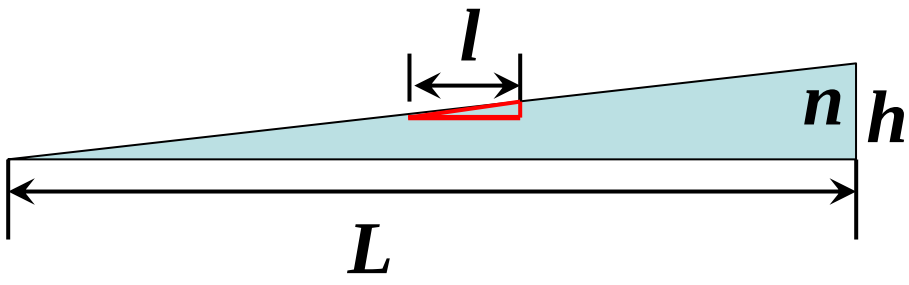
$$l = \frac{d_{k+1} - d_k}{\sin\theta} \approx \frac{\lambda}{2n\theta}$$

θ 增大条纹向劈尖棱边移动，间距变小

若向上移动上玻璃片，条纹又如何变化？

- 劈尖膜干涉的应用

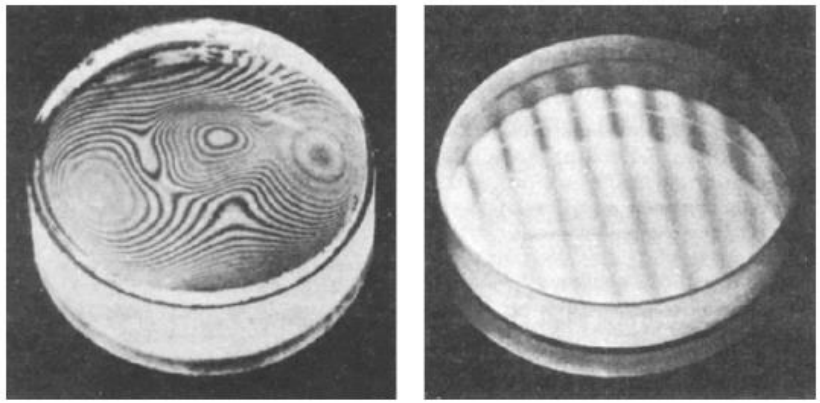
$$l = \frac{\lambda}{2n\theta}$$

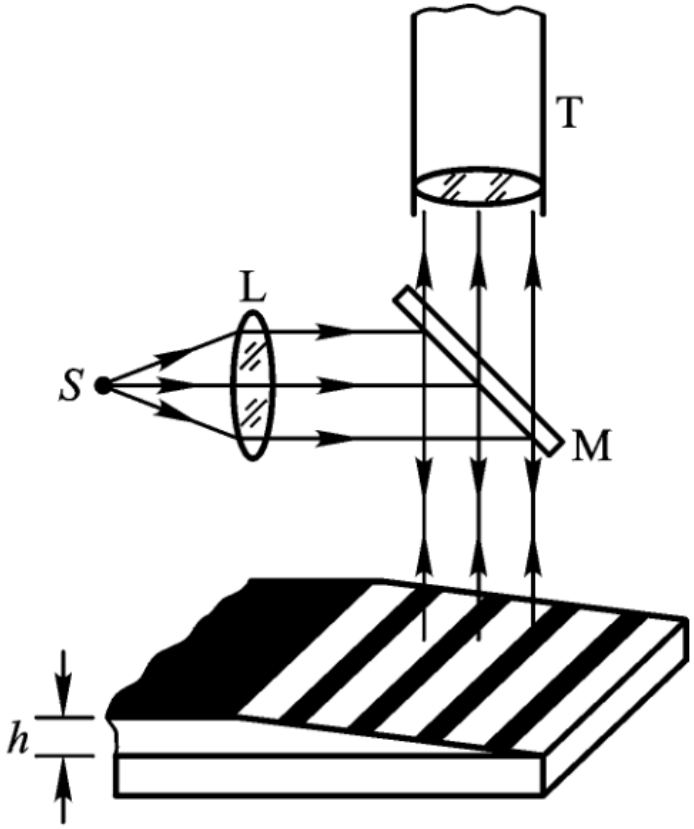


若劈尖夹角已知，通过测量条纹间距可测波长

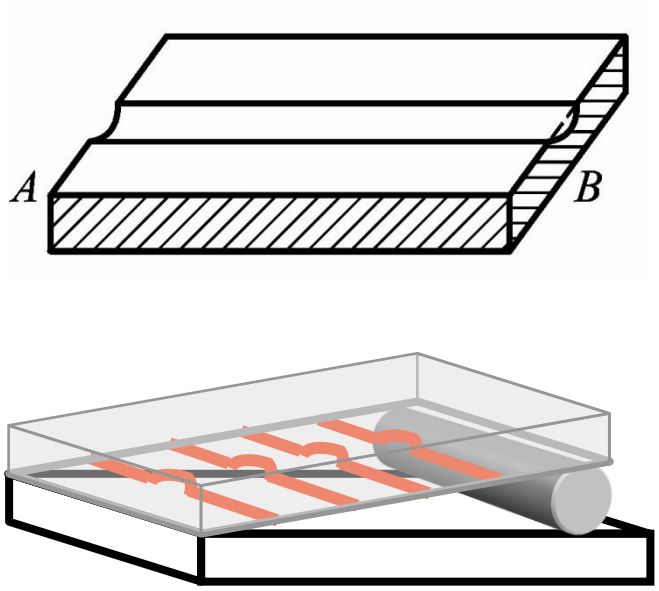
若单色光波长已知可以测量微小直径、厚度

还可测量弓箭表面平整度



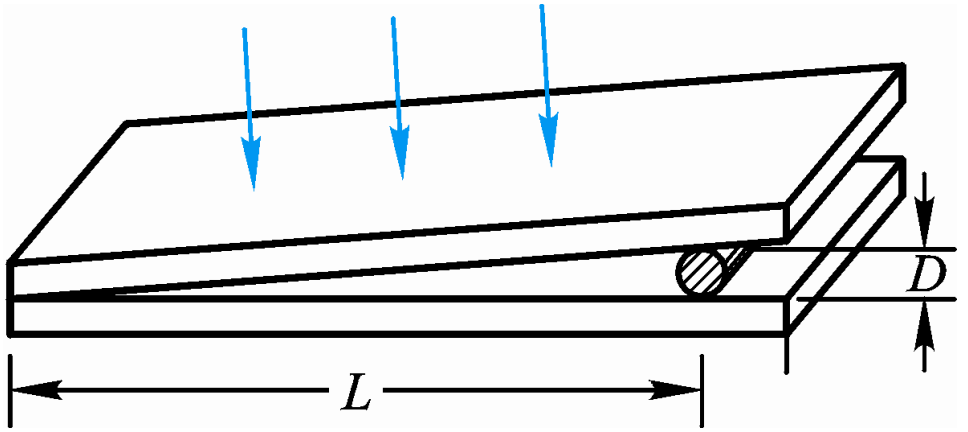


测量SiO_2薄膜厚度



空气劈尖测表面沟纹

例 为了测量金属细丝的直径，把金属丝夹在两块平玻璃之间，形成劈尖，如图所示，如用单色光垂直照射，就得到等厚干涉条纹。测出干涉条纹的间距，就可以算出金属丝的直径。某次的测量结果为：单色光的波长λ =589.3 nm金属丝与劈间顶点间的距离 L=28.880 mm，30条明纹间得距离为4.295 mm,求金属丝的直径D?



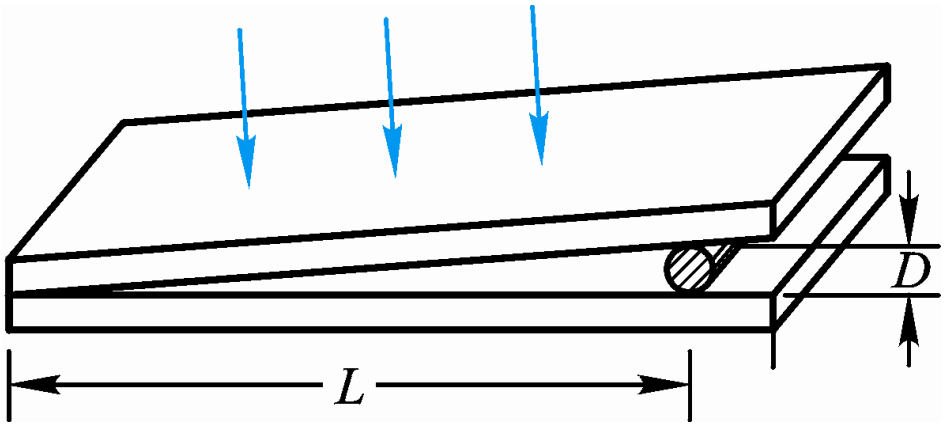
解： 相邻两条明纹间的间距

$$l = \frac{4.295}{29}\ \text{mm}$$

$$l\sin\theta = \frac{\lambda}{2} \qquad \sin\theta \approx \frac{D}{L}$$

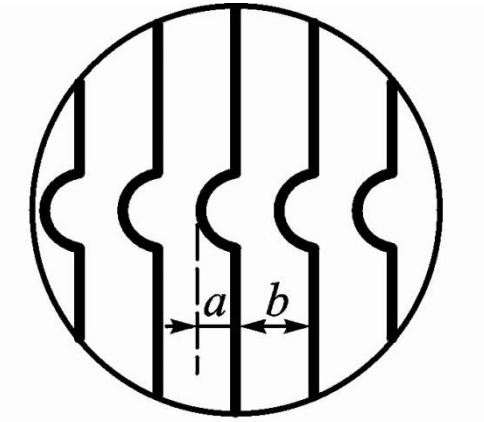
$$\Rightarrow \quad D = \frac{L}{l}\frac{\lambda}{2}$$

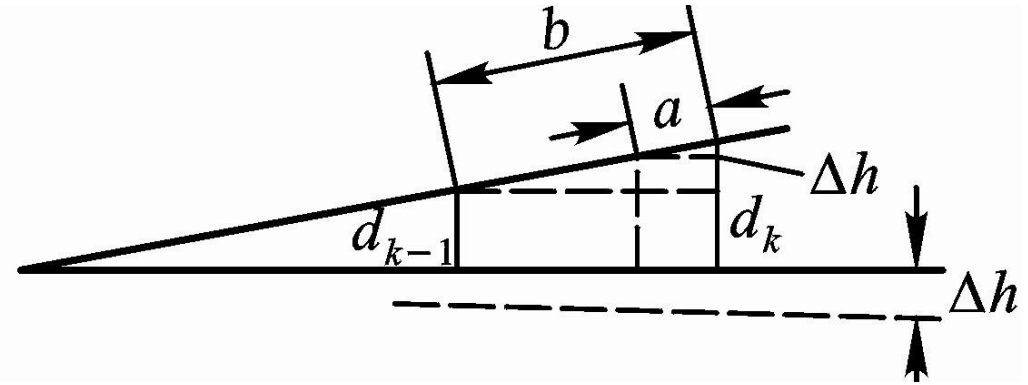
$$D = 0.05746\ \text{mm}$$



例 利用空气劈尖的等厚干涉条纹可以检测工 件表面存在的极小的加工纹路， 在经过精密加工的工件表面上放一光学平面玻璃，使其间形成空气劈形膜，用单色光照射玻璃表面，并在显微镜下观察到干涉条纹，如图所示。试根据干涉条纹的弯曲方向，判断工件表面是凹的还是凸的；并证明凹凸深度可用下式求得

$$\Delta h = \frac{a}{b}\frac{\lambda}{2}$$

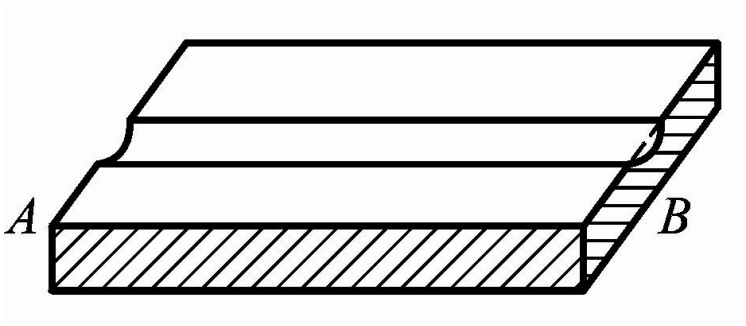




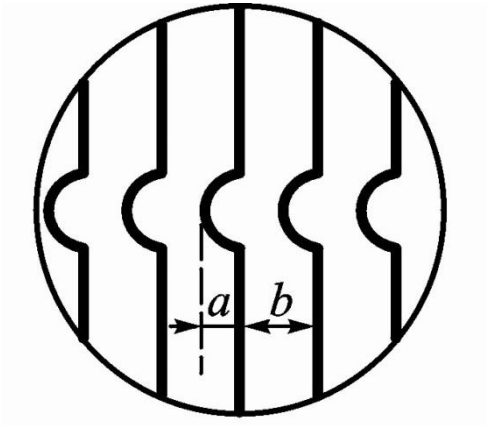
解： 观察到的干涉条纹弯向空气膜的左端，可判断工件表面是下凹的，如图所示。

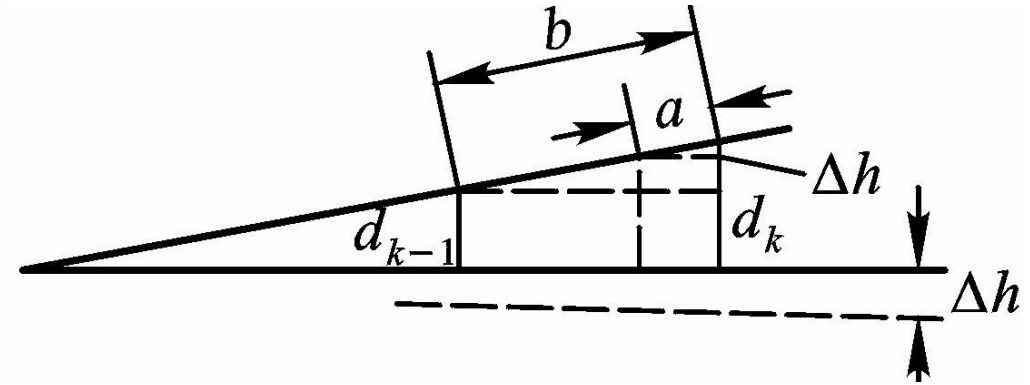
由图中相似直角三角形：

$$\frac{a}{b}=\frac{\Delta h}{(d_k-d_{k-1})}=\frac{\Delta h}{\frac{\lambda}{2}}$$



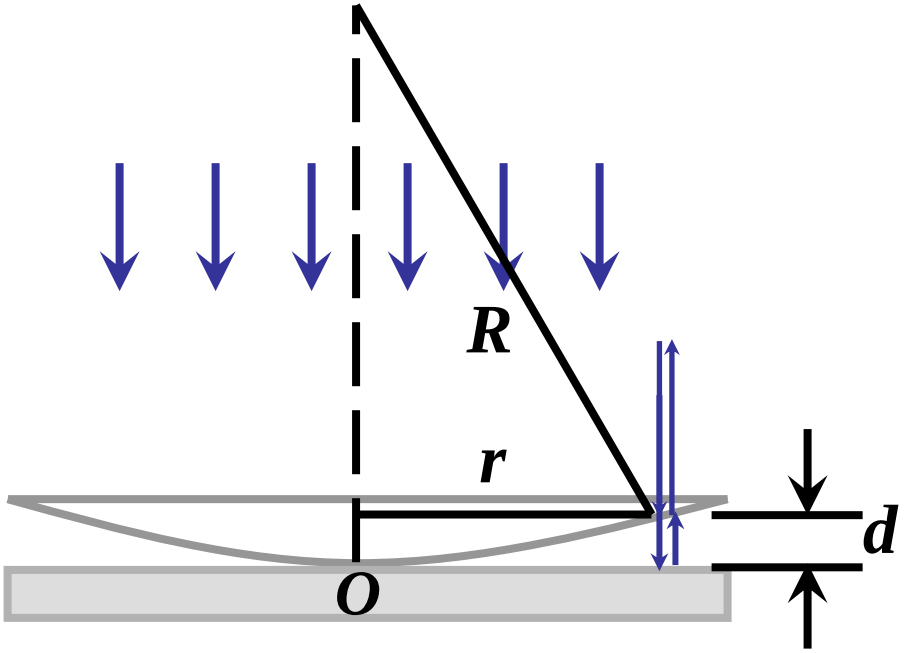
$$\Rightarrow \quad \Delta h=\frac{a}{b}\frac{\lambda}{2}$$





2、牛顿环

平行光垂直入射到平凸透镜
与平玻璃表面之间的空气膜。



$$\delta = 2d + \frac{\lambda}{2} = \begin{cases} k\lambda \\ (2k+1)\frac{\lambda}{2} \end{cases}$$

由几何关系

$$r^2 = R^2 - (R-d)^2 = 2Rd - d^2 \approx 2Rd \Rightarrow d = \frac{r^2}{2R}$$

$$r_{明} = \sqrt{\frac{(2k-1)R\lambda}{2}} \qquad r_{暗} = \sqrt{kR\lambda}$$

- 条纹特点

形状：同心圆环（由等厚条纹+几何结构决定）

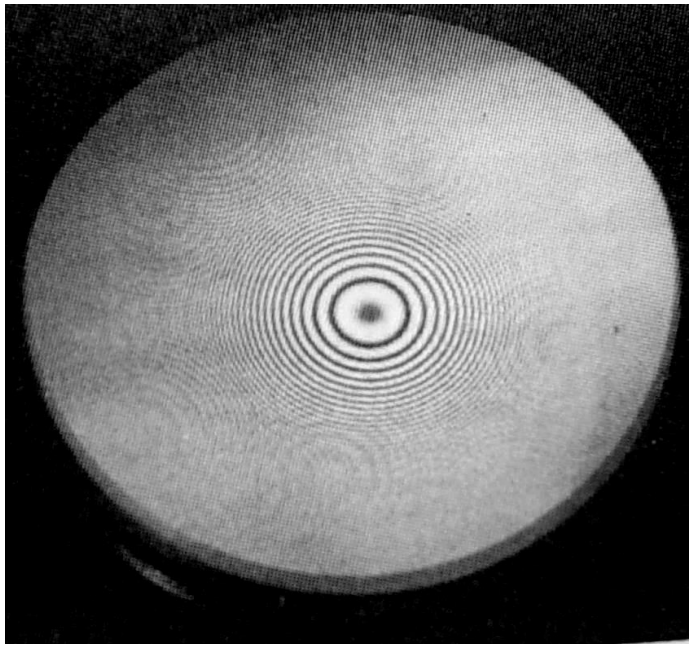
条纹级次分布：　圆心处为0级暗纹。

半径越大，级次越高。(与等倾干涉条纹不同)

若压紧透镜，牛顿环的条纹向外扩张。

条纹间隔分布：内疏外密

$$r_{明}=\sqrt{\frac{(2k-1)R\lambda}{2}}$$



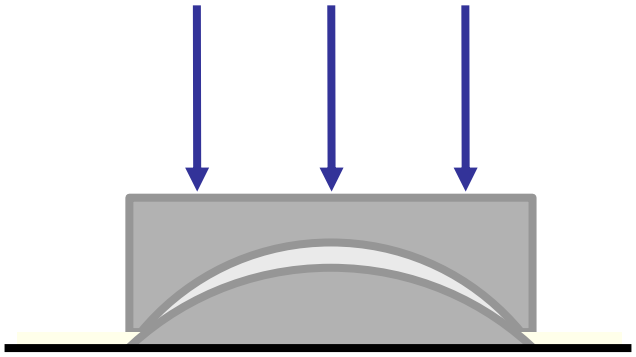
- 牛顿环的应用

$$r_{暗}=\sqrt{\frac{kR\lambda}{n}}$$

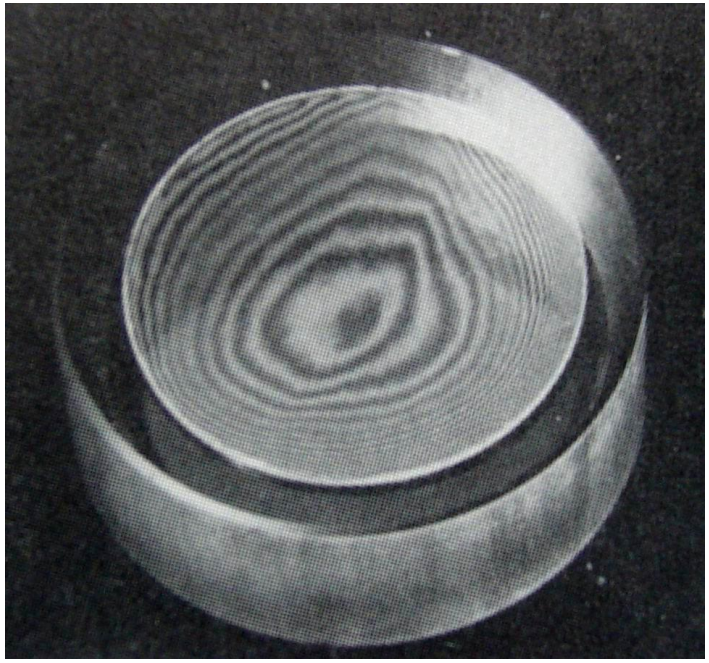
$$\Rightarrow \quad r_{k+m}^2-r_k^2=m\frac{R\lambda}{n}$$

测量第$k+m$和第k级暗条纹半径

⇨ 波长或球面透镜半径



检验透镜表面质量



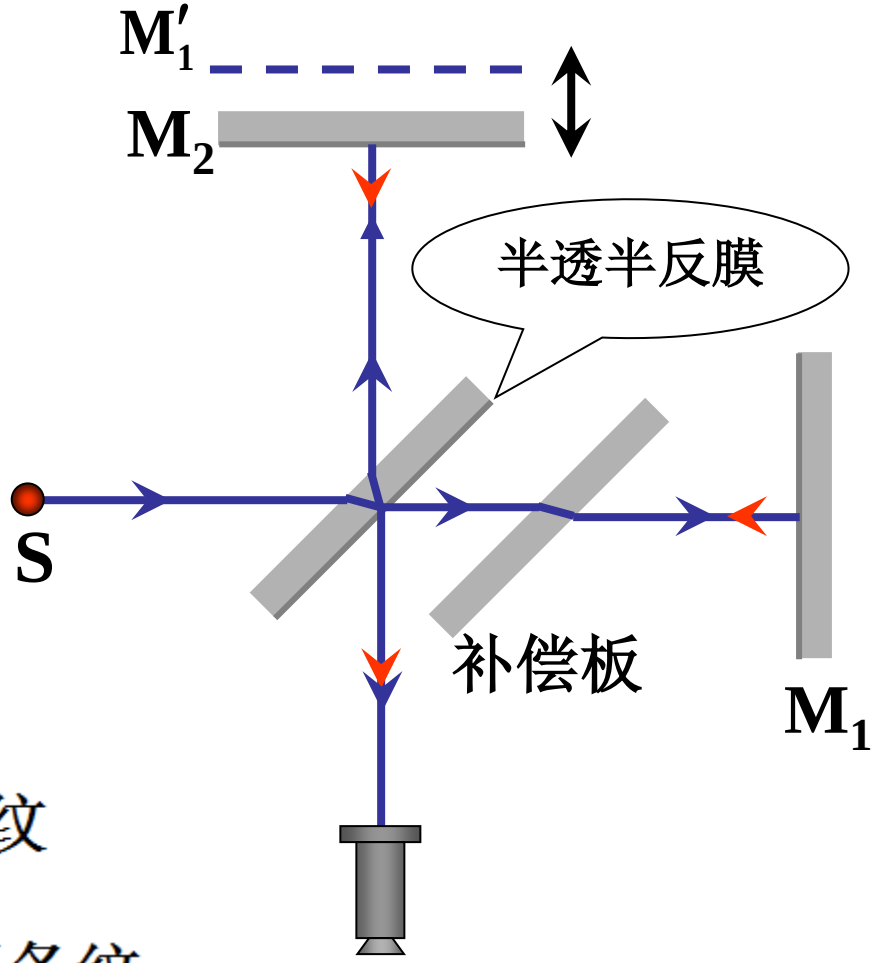
§1-5 迈克耳孙干涉仪

迈克耳孙干涉仪 (1880年)

1. 结构

半透半反膜

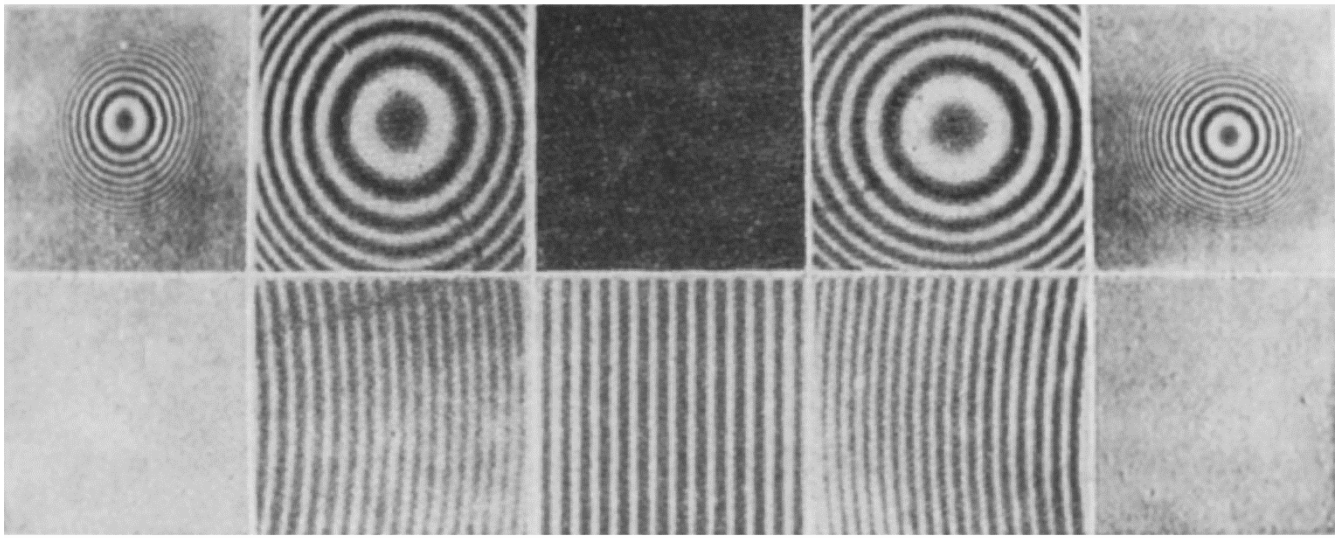
补偿板

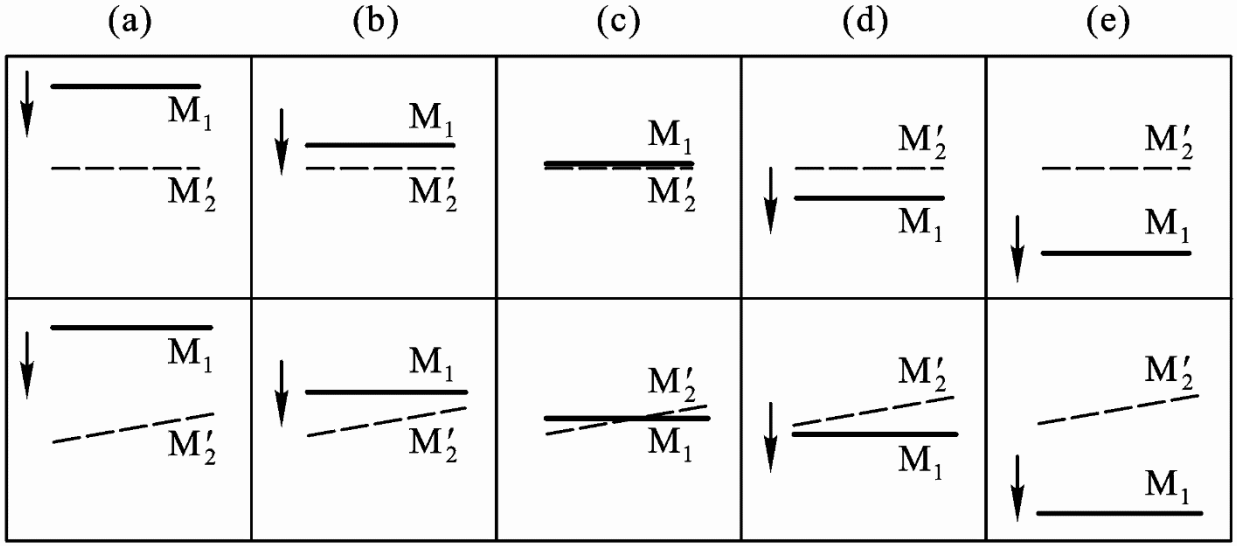


2. 原理

M_1、M_2垂直 ⇨ 等倾条纹

M_1、M_2偏离垂直 ⇨ 等厚条纹


(a)
(b)
(c)
(d)
(e)
(f)
(g)
(h)
(i)
(j)


(a)
(b)
(c)
(d)
(e)
M1
M′2
M1
M′2
M1
M′2
M′2
M1
M′2
M1
M1
M′2
M1
M′2
M′2
M1
M′2
M1
M′2
M1
(f)
(g)
(h)
(i)
(j)

3. 应用

• 若M_2平移Δd 时，干涉条纹移过N条，则有

$$\Delta\delta = 2\Delta d = N\lambda \quad \Rightarrow \quad \Delta d = N \cdot \frac{\lambda}{2}$$

由此可测量微小位移Δd 或波长。

• 条纹的移动也可由其他原因引起，如介质膜的插入或移去，此时引起的条纹移动数目：

$$N = 2(n-1)d / \lambda$$

由此可测定介质膜的厚度 d 或折射率 n 。

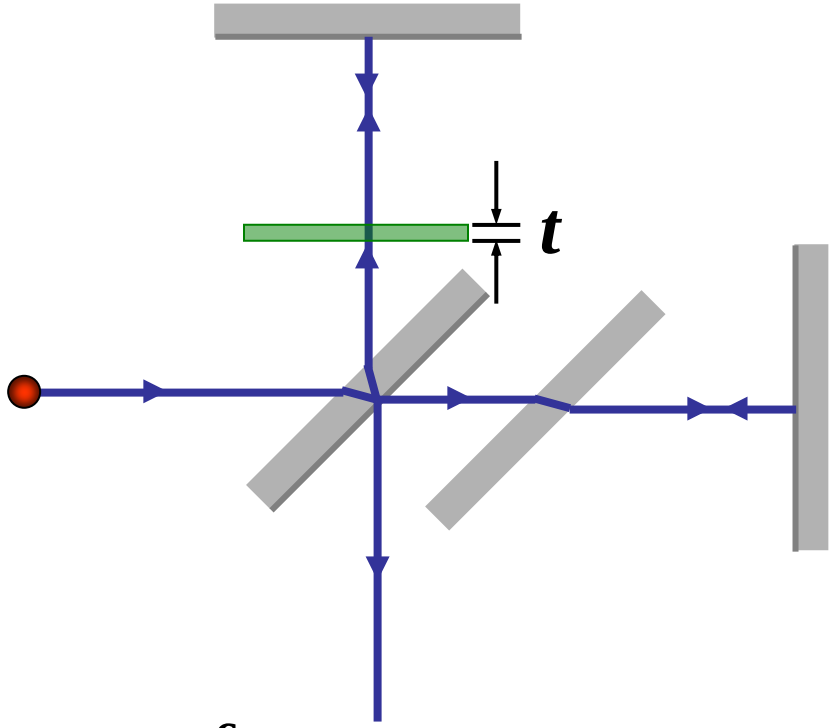
• 可测定光谱的精细结构。

思考： 当把折射率为n=1.40的薄膜放入迈克耳孙干涉仪的一臂时，如果产生了7.0条条纹的移动，确定薄膜的厚度。（已知钠光的波长为λ = 589.3 nm）

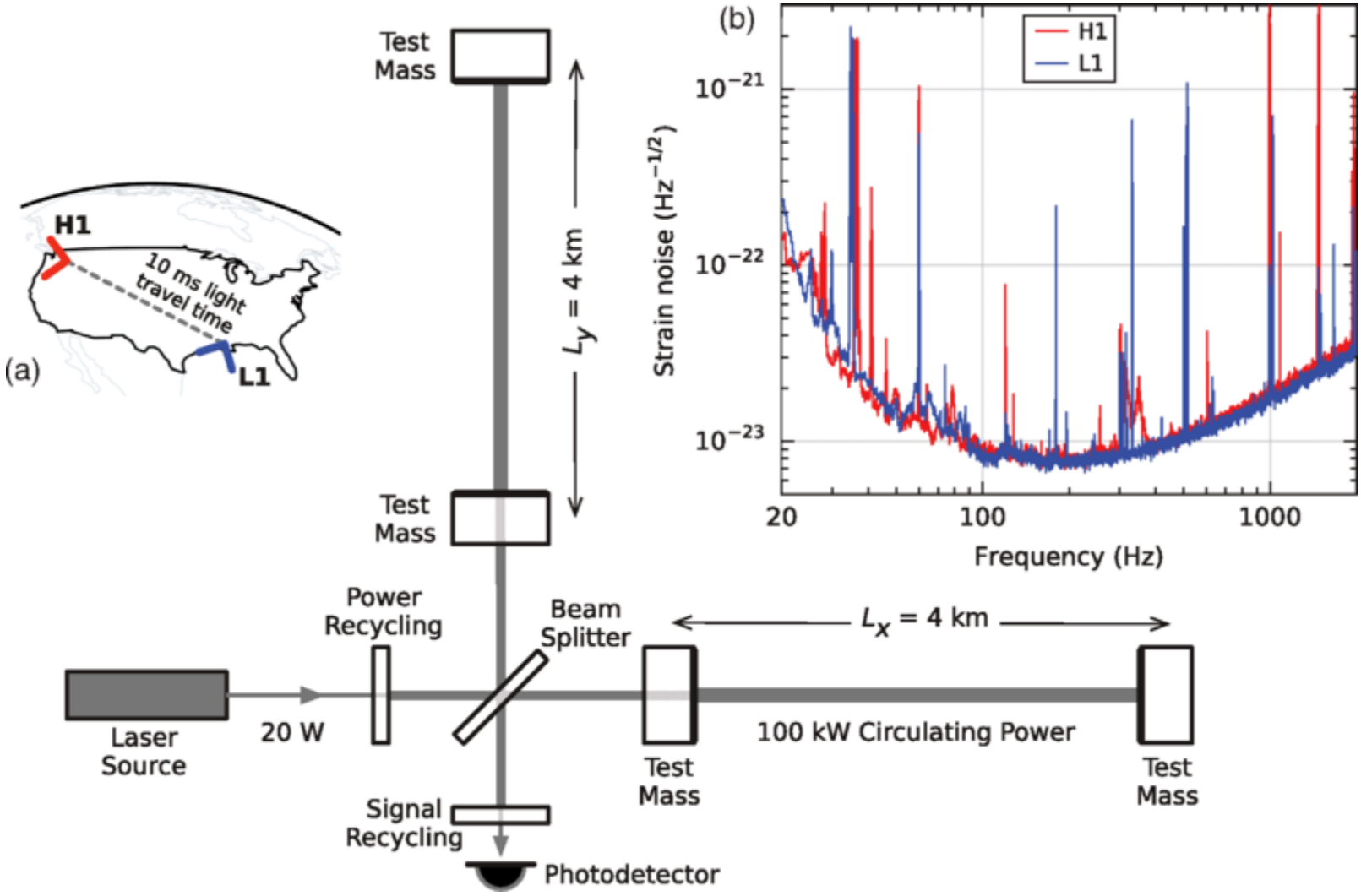
$$\Delta\delta = 2(n-1)t = \Delta k\lambda$$

$$t = \frac{\Delta k \cdot \lambda}{2(n-1)}$$

$$= \frac{7\times 589.3\times 10^{-9}}{2(1.4-1)}\,\mathrm{m} = 5.154\times^{-6}\,\mathrm{m}$$



LIGO引力波探测: h~10^-22



图片来源：wiki

不做要求

1.6 Fabry-Perot干涉仪 多光束干涉

(1) 问题的提出（双光束→多光束）

Michelson干涉仪是应用分振幅原理的干涉仪，它是一个双光束系统，其干涉图样的强度分布函数为 $4A_1^2\cos^2\frac{\Delta\varphi}{2}$，在实际应用中，力求图样是狭窄、清晰、明亮的条纹。

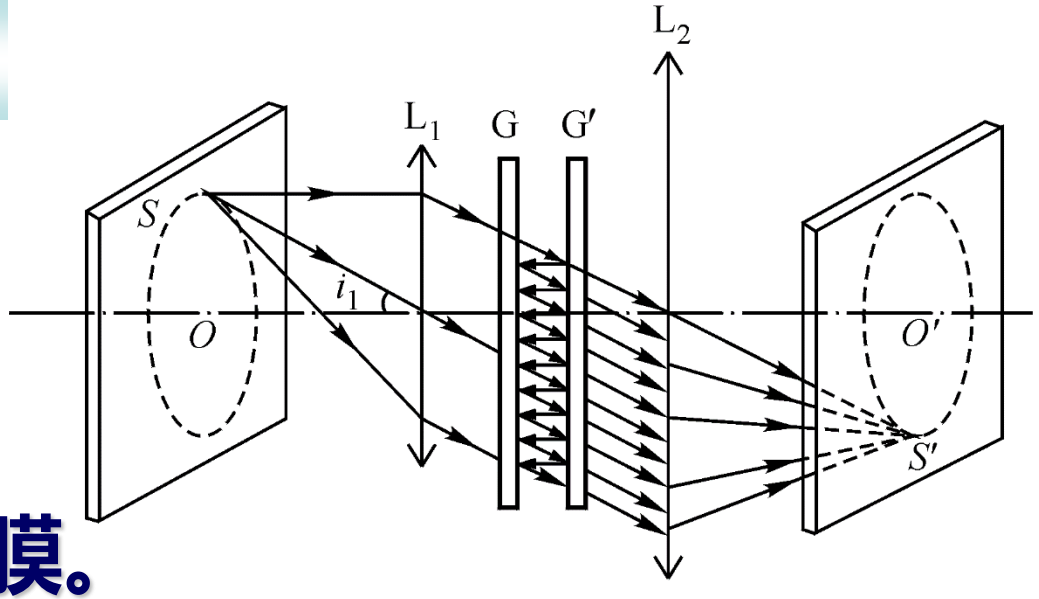
采用相位差相同的多光束系统可实现。典型装置为Fabry-Perot干涉仪。

来自：姚启钧，光学教程 课件

Fabry-Perot干涉仪 多光束干涉

(2) Fabry-Perot干涉仪

- 装置



平行放置的两块平板G和G′组成。 G、G ′相向的表面镀有薄银膜。

面光源 S 放在透镜L_1的焦平面上。则许多方向不同的平行光束入射到干涉仪上，经G、G ′间多次来回反射，最后透射出来的平行光束在透镜 L_2 的焦平面上形成同心圆形的明锐等倾干涉条纹。

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(2) Fabry-Perot干涉仪

- 干涉图样

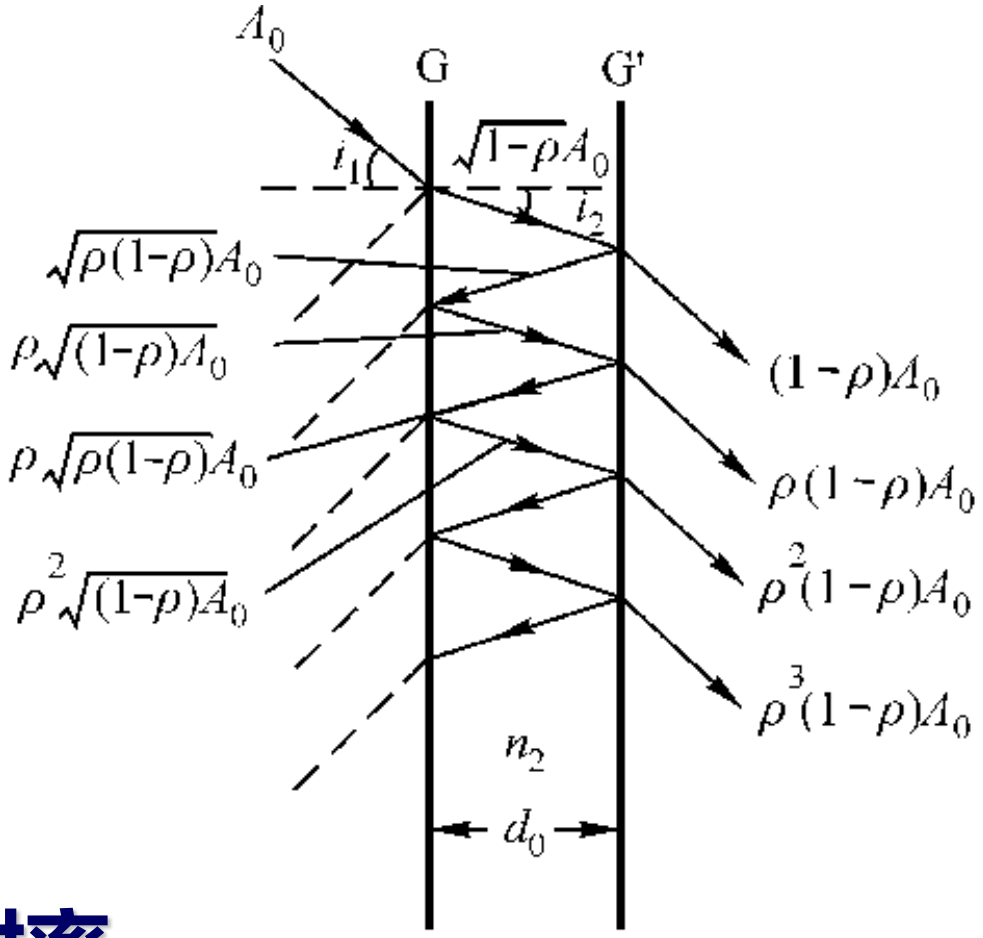
从G′后表面相继透射出来的平行光束的振幅依次为

$(1-\rho)A_0$

$\rho(1-\rho)A_0$

$\rho^2(1-\rho)A_0$

式中ρ为镀银面的反射率。



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(2) Fabry-Perot干涉仪

相邻每两束光在到达L_2的焦平面上的同一时，彼此的光程差均相等。

$$\delta = 2n_2 d_0 \cos i_2$$

相应的相位差为 $\phi = \frac{4\pi}{\lambda} n_2 d_0 \cos i_2$

各光束的相位依次为0, φ , 2φ ,…

多束光叠加的合振幅为 $A^2 = \dfrac{A_0^{\ 2}}{1 + \dfrac{4\rho}{(1-\rho)^2} \sin^2 \dfrac{\varphi}{2}}$

Fabry-Perot干涉仪 多光束干涉

(2) Fabry-Perot干涉仪

强度的讨论（Airy函数）

ρ 给定，则 $A^2=\dfrac{A_0^{\,2}}{1+\dfrac{4\rho}{(1-\rho)^2}\sin^2\dfrac{\varphi}{2}}=A^2(\varphi)$

$A_{\max}^{\,2}=A_0^{\,2}\qquad \varphi=0,2\pi,4\pi,\cdots$

$A_{\min}^{\,2}=A_0^{\,2}\left[1+\dfrac{4\rho}{(1-\rho)^2}\right]^{-1}\qquad \varphi=\pi,3\pi,5\pi,\cdots$

$\rho\to 0$ ，A^2几乎不变。

$\rho\to 1$ ，只有 $\varphi=0,2\pi,4\pi,\cdots$ 时出现最大值。

Fabry-Perot干涉仪 多光束干涉

(2) Fabry-Perot干涉仪

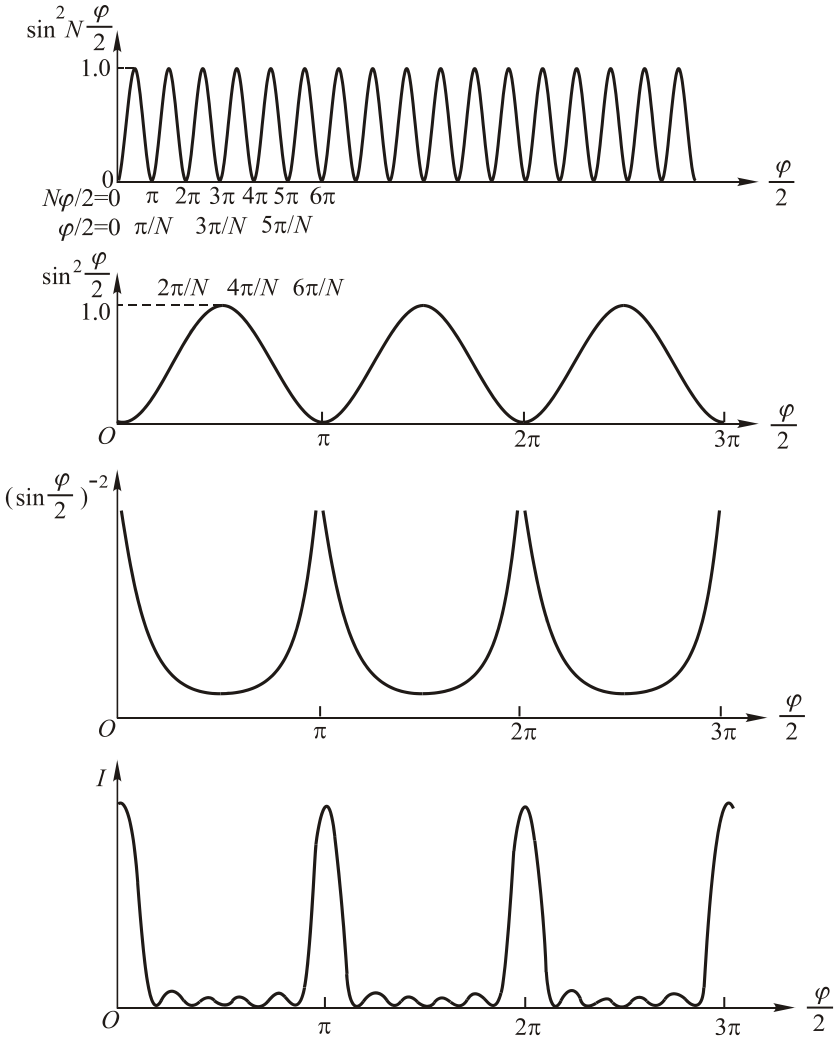
此时透射光干涉图样由全黑的背景下一组很细的亮条纹构成，即条纹的锐度和可见度很大。这接近于**等振幅的多光束干涉**。叠加结果的合振幅为

$$A^2 = A_0^{\,2}\frac{\sin^2\frac{N\varphi}{2}}{\sin^2\frac{\varphi}{2}}$$

N为光束的总数，φ则为各相邻光束之间的相位差。

Mathematica展示多光束干涉函数图形

教材图1-27



干涉小结

- 光是电磁波。电磁波具有从射电到可见光，到X-ray以及更高能量的连续电磁波谱。
- 相干条件：振动方向相同，频率相同，具有确定的相位差。可以通过分波阵面法或分振幅法获得相关光。
- 杨氏双缝干涉、薄膜干涉（包括牛顿环，迈克尔孙干涉仪等），光学三问：
  （1）光程差，（2）条纹位置，（3）条纹间距。
- 半波损失：具体问题具体分析

Review & Summary (Halliday)

Huygens' Principle The three-dimensional transmission of waves, including light, may often be predicted by *Huygens' principle,* which states that all points on a wavefront serve as point sources of spherical secondary wavelets. After a time t, the new position of the wavefront will be that of a surface tangent to these secondary wavelets.

The law of refraction can be derived from Huygens' principle by assuming that the index of refraction of any medium is $n = c/v$, in which v is the speed of light in the medium and c is the speed of light in vacuum.

Wavelength and Index of Refraction The wavelength λ_n of light in a medium depends on the index of refraction n of the medium:

$$\lambda_n = \frac{\lambda}{n}, \qquad (35\text{-}6)$$

in which λ is the wavelength in vacuum. Because of this dependency, the phase difference between two waves can change if they pass through different materials with different indexes of refraction.

双语学习用

Young's Experiment In **Young's interference experiment,** light passing through a single slit falls on two slits in a screen. The light leaving these slits flares out (by diffraction), and interference occurs in the region beyond the screen. A fringe pattern, due to the interference, forms on a viewing screen.

The light intensity at any point on the viewing screen depends in part on the difference in the path lengths from the slits to that point. If this difference is an integer number of wavelengths, the waves interfere constructively and an intensity maximum results. If it is an odd number of half-wavelengths, there is destructive interference and an intensity minimum occurs. The conditions for maximum and minimum intensity are

$$d \sin \theta = m\lambda, \qquad \text{for } m = 0, 1, 2, \ldots$$
$$\text{(maxima—bright fringes)}, \tag{35-14}$$

$$d \sin \theta = (m + \tfrac{1}{2})\lambda, \qquad \text{for } m = 0, 1, 2, \ldots$$
$$\text{(minima—dark fringes)}, \tag{35-16}$$

where θ is the angle the light path makes with a central axis and d is the slit separation.

Coherence If two light waves that meet at a point are to interfere perceptibly, the phase difference between them must remain constant with time; that is, the waves must be **coherent.** When two coherent waves meet, the resulting intensity may be found by using phasors.

Intensity in Two-Slit Interference In Young's interference experiment, two waves, each with intensity I_0, yield a resultant wave of intensity I at the viewing screen, with

$$I = 4I_0 \cos^2 \tfrac{1}{2}\phi, \qquad \text{where } \phi = \frac{2\pi d}{\lambda} \sin \theta. \qquad (35\text{-}22, 35\text{-}23)$$

Equations 35-14 and 35-16, which identify the positions of the fringe maxima and minima, are contained within this relation.

Thin-Film Interference When light is incident on a thin transparent film, the light waves reflected from the front and back surfaces interfere. For near-normal incidence, the wavelength conditions for maximum and minimum intensity of the light reflected from a *film in air* are

$$2L = (m + \tfrac{1}{2})\frac{\lambda}{n_2}, \qquad \text{for } m = 0, 1, 2, \ldots$$

(maxima—bright film in air), (35-36)

$$2L = m\frac{\lambda}{n_2}, \qquad \text{for } m = 0, 1, 2, \ldots$$

(minima—dark film in air), (35-37)

where n_2 is the index of refraction of the film, L is its thickness, and λ is the wavelength of the light in air.

If the light incident at an interface between media with different indexes of refraction is in the medium with the smaller index of refraction, the reflection causes a phase change of π rad, or half a wavelength, in the reflected wave. Otherwise, there is no phase change due to the reflection. Refraction causes no phase shift.

The Michelson Interferometer In *Michelson's interferometer* a light wave is split into two beams that, after traversing paths of different lengths, are recombined so they interfere and form a fringe pattern. Varying the path length of one of the beams allows distances to be accurately expressed in terms of wavelengths of light, by counting the number of fringes through which the pattern shifts because of the change.